



Research on interaction optimization methods for AR games in autonomous driving: Based on contextual awareness and virtual avatar design[☆]

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ARTICLE INFO

Keywords:

Smart cockpit
Multimodal game design
Situational awareness
Autonomous driving
Artificial intelligence

ABSTRACT

With the continuous development of artificial intelligence and autonomous driving technology, cars are transitioning from traditional transportation to user-centered intelligent mobile spaces. In order to enhance drivers' road safety awareness and entertainment experience, this study explores the design of multi-modal games in smart cockpits in the context of L3-L4 autonomous driving based on the theory of Situation Awareness (SA). First, using SA and KANO models, key scenarios and design elements are identified, and a method combining context-awareness and avatar design is proposed. Second, typical driving scenarios are identified through user interviews and evaluation scales, and experimental studies are conducted to analyze and refine the multi-modal game interaction information in the smart cockpit environment. Finally, based on Pokémon-style, an AR game design solution suitable for a smart cockpit was developed. The 2 (with and without game interaction) × 4 (vision, vision + hearing, vision + touch, vision + hearing + touch) mixed factorial experimental design was designed, and the proposed design scheme was evaluated using a driving simulator. The experimental results demonstrate that the game interaction design for L3-L4 autonomous driving conditions significantly enhances driver perception of road and environmental information, reduces reaction time in emergencies, improves user experience, and increases trust in the autonomous driving system compared to the non-game interaction design.

1. Introduction

With the rapid advancement of artificial intelligence (AI) and autonomous driving technologies, vehicles are evolving from traditional transportation tools into intelligent, user-centered mobile spaces [1]. Particularly at the L3-L4 levels of automation, the driving role shifts toward passive supervision, allowing users to engage in secondary tasks such as entertainment and social interaction [2]. In response, the intelligent cockpit has emerged as a critical platform that integrates in-car information systems, driver monitoring, and multimodal human-machine interfaces (HMI) to support both safety and user experience.

As drivers transition into passengers during autonomous driving, the demand for immersive in-car entertainment—especially interactive games—has increased significantly. This shift raises a fundamental challenge: how to design engaging in-car entertainment systems that enhance the user experience without compromising driving safety or

situational awareness [3]. Situation Awareness (SA) theory, widely used in high-risk domains such as aviation and military operations, offers a valuable framework for addressing this challenge. Comprising three levels—perception, comprehension, and projection—SA theory supports the evaluation and enhancement of users' understanding of dynamic environments. While SA has been applied to the design of driver assistance systems and vehicle interfaces, its application in in-car game design remains limited. Most existing systems focus on screen-based passenger entertainment, lacking real-time integration with vehicle and road data [4].

Recent developments in Augmented Reality (AR) and multimodal interaction technologies, including voice, gesture, eye-tracking, and haptic feedback, offer new possibilities for interactive game experiences within intelligent cockpits. These technologies can overlay virtual content on real-world environments, enabling situationally-aware, personalized, and immersive game interactions. However, the key challenge

[☆] This paper was recommended for publication by Prof Guangtao Zhai.

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<https://doi.org/10.1016/j.displa.2025.103180>

Received 14 April 2025; Received in revised form 17 July 2025; Accepted 11 August 2025

Available online 12 August 2025

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remains: how can these interactions be designed to support driver awareness and safety, especially under L4 autonomous driving conditions?

This study combines SA theory with AR game design and proposes a set of multimodal game interaction methods for smart cockpits in the context of autonomous driving. (1) At the level of theory integration, SA theory is introduced from traditional aviation, military, human factors engineering and other high-risk control fields to the intelligent cockpit environment, and the three-phase model of “perception-comprehension-prediction” is used to guide the construction of the game interaction logic, which realizes the theoretical support for the integration of entertainment and safety. (2) At the interaction modeling level, the system integrates vision, hearing and touch multimodal interaction channels, and maps the SA theoretical structure to the game’s visual cue design, virtual character behavioral feedback, and task-driven mechanism, to construct an immersive Pokémon-style game interaction process that is environmentally sensitive, informationally responsive, and decision-making-guided, so that the user can still maintain cognitive state of key elements of the road under the state of autopilot. maintain the cognitive state of the key elements of the road. Finally, at the evaluation and validation level, an experimental platform is constructed based on a driving simulator, combining eye tracking, reaction time, NASA-TLX workload scale and subjective trust score and other multivariate evaluation tools to quantitatively analyze the perceptual effects and cognitive loads of the interaction strategy, validate the framework’s effectiveness in improving the road information perception efficiency and the user interaction experience, and put forward a data-driven optimization path based on the user’s feedback. A data-driven optimization path is proposed, which provides a scalable reference paradigm for AR interaction design in future smart cockpits.

2. Related research

2.1. Human-vehicle collaboration in intelligent cockpits

With the widespread penetration of intelligent technologies in the automotive sector, research into intelligent cockpits has become a key direction for the integrated development of human-machine interaction and autonomous driving. Intelligent cockpits have broken through the boundaries of traditional vehicle interior design, integrating automated control and human-machine interaction (HMI) technologies to create a user-centric, highly perceptive mobile experience [5]. Current mainstream systems are equipped with networked and AI-driven components such as digital instrument clusters, infotainment systems, HUD head-up displays, and rear-seat entertainment screens [6,7], which achieve synergy between driving control, user interaction, and environmental perception through a unified HMI framework [8]. The core evolutionary trend of intelligent cockpits is the shift from responsive to predictive interaction, leveraging artificial intelligence, affective computing, and user modelling technologies to achieve real-time insights into and dynamic responses to the driver’s state [9].

Building on this, some scholars have begun to propose diverse human-vehicle collaboration mechanisms to optimise cabin interaction experiences. For example, Reference [10] designed an intelligent carpet for the cockpit, integrating sponge-type triboelectric sensors to detect driver foot movements, enabling low-interference human-vehicle interaction intent recognition and effectively reducing visual load and hand operation pressure. Literature [11] draws on human-machine hybrid intelligence, introducing the ‘schooling effect’ model to construct a highly adaptive and scalable multi-vehicle collaborative intelligent driving system, demonstrating the advantages of collective intelligence in cross-vehicle collaboration. From a control strategy perspective, literature [12] constructs a collaborative driving framework integrating human driver and co-pilot control inputs, validated on a test platform supporting manual/automatic switching. This framework emphasises real-time response capabilities based on internal and external risk

perception. Furthermore, Reference [13] systematically reviews the ‘human-vehicle trust-based collaboration’ approach for autonomous driving, proposing a brain-inspired intelligent cognitive model focused on driver behaviour understanding and cognitive decision planning. It also highlights current challenges such as explainability, trust establishment, and system acceptance. In terms of communication and environmental modelling, Reference [14] explores a cooperative driving model in a 6G environment, proposing a system state prediction and safe following distance mapping method based on Markov chains, effectively improving speed control smoothness and traffic flow efficiency. The study validated the model’s robustness and potential for improving traffic efficiency through simulation on the AirSim platform.

In summary, existing research has conducted in-depth explorations in areas such as perception fusion, control coordination, cognitive modelling, and communication optimisation, clarifying the technological development path for intelligent cockpits and human-vehicle collaborative interaction. However, most work remains focused on system functional and structural optimisation, lacking dynamic interaction mechanism modelling oriented towards user experience. Especially in high cognitive load scenarios, such as night driving or foggy weather conditions, active regulation mechanisms for users’ situational awareness capabilities and emotional state changes are still underdeveloped. Additionally, current systems largely lack empirical research support for interaction behaviours, making it difficult to effectively evaluate the actual impact of complex human-vehicle collaborative interaction strategies on driving safety and immersion. Therefore, this paper proposes a multi-modal intelligent interaction design method that integrates gamification mechanisms. Through virtual rewards, dynamic prompts, and multi-channel perception enhancement techniques, this method strengthens user participation motivation and situational response capabilities. This method not only enhances the interaction adaptability of the cockpit system but also provides a new path for constructing intelligent interaction models with active intent recognition and situational awareness capabilities.

2.2. Multimodal interaction technologies in cockpit systems

Multimodal interaction technology, as a key enabler of intelligent cockpits, is increasingly expanding beyond functional control to more complex human-machine adaptation tasks such as emotion recognition, personalised services, and proactive interaction. Currently, channels such as visual (e.g., facial recognition, gesture recognition), auditory (voice commands, spatial sound effects), tactile (vibration feedback, environmental touch control), and physiological sensing (eye tracking, heart rate, electroencephalography) are being progressively integrated into cockpit systems, forming a ‘perception-cognition-response’ closed-loop human-vehicle collaborative interaction mechanism [15,16]. Traditional central control buttons and touchscreen interaction methods are gradually being replaced by integrated solutions combining voice, vision, and haptic feedback, which are being applied to new scenarios such as driving assistance, emotional regulation, identity verification, and multi-modal task allocation, as shown in Fig. 1.

In future trends, intelligent cockpits will evolve toward ‘decentralised control,’ ‘context-aware driving,’ and ‘emotionally companionable services.’ Technological advancements represented by multimodal large language models (MLLM) are driving voice assistants to transition from simple command execution to proactive companion systems based on emotional recognition. Environmental surfaces such as windows and seats are emerging as potential interaction media, while physiological signals like eye tracking and heart rate changes are being used for dynamic emotional regulation and stress intervention [17]. In AR in-vehicle gaming scenarios, multimodal interaction not only enhances immersion and engagement but also supports dynamic adjustment of interaction intensity based on emotional states, creating a more context-adaptive digital space [18].

Multiple empirical studies have further revealed the applicability

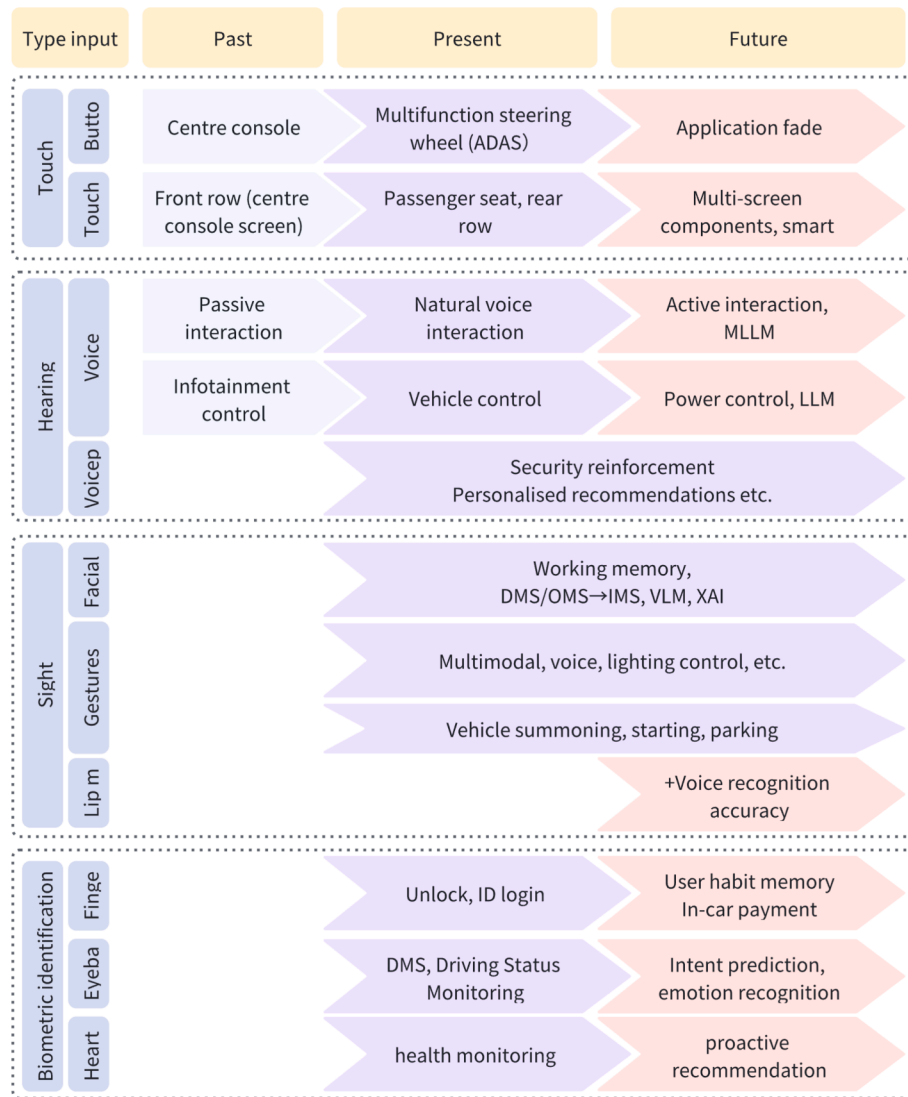


Fig. 1. Trends in multimodal interaction technology.

and challenges of multimodal interaction technology. For example: Literature [19] conducted a survey analysis of the adaptability differences among three cockpit interaction solutions (simulated switches, gesture/voice control, touchscreen) among users, indicating that individual user preferences significantly influence the evaluation of multimodal solutions. Literature [20] designed a multimodal interaction framework for emotional regulation, using a 2×3 mixed experimental design to validate the positive effects of modality combinations such as auditory-visual and auditory-olfactory on anger relief. Literature [21] explored the impact of visual, auditory, and vibrational modalities on immersion and comfort assessments, finding that visual interventions have a low marginal effect, suggesting that future experimental designs may appropriately weaken their role. Literature [22] proposed the GLM-TripleGen model, which uses a large language model to construct an intelligent cockpit knowledge graph, enabling the exploration of complex relationships between states and behaviours. Literature [23] assessed cognitive stress in different noise environments using eye movements, pupil changes, and NASA-TLX cognitive load scores, verifying the negative impact of speech recognition errors on interaction performance in noisy environments. Literature [24] combines object detection algorithms with explainable AI, significantly improving interaction response speed and accuracy in mixed reality systems. Literature [25–27] proposes multimodal fusion methods based on

machine learning and deep learning in fields such as IoT-driven fertiliser recommendations, crowd anomaly detection, and Alzheimer’s disease diagnosis, indirectly corroborating the transferability and generalisability of multimodal technology in high-complexity perception tasks.

Although current research has made substantial progress in the functional implementation and perception fusion pathways of multimodal interaction technology, most studies focus on technical feasibility verification in single scenarios, lacking systematic modelling and quantitative assessment of dynamic cognitive states, interaction loads, and emotional feedback mechanisms in complex driving environments. Especially in real driving scenarios, constrained by interfering environments (such as noise, lighting, rain, and snow), individual user differences, and operational redundancy, achieving real-time, stable, and personalised multimodal interaction remains a core challenge. Additionally, existing literature primarily evaluates systems based on ‘system capabilities’ or ‘perceptual accuracy,’ neglecting the dynamic evolution of subjective experiences, cognitive load, and behavioural responses during the interaction process. This omission prevents multimodal interaction systems from supporting the transition from the ‘operational layer’ to the ‘cognitive layer’ and ‘emotional layer.’

To address this gap, this paper innovatively proposes a multimodal contextual interaction model based on AR game mechanisms, integrating situational awareness theory and emotional feedback

mechanisms. Through a closed-loop interactive logic of ‘task triggering – cognitive regulation – immediate reward,’ the model aims to enhance users’ situational awareness capabilities and interaction response efficiency in L4 autonomous driving environments. This model not only focuses on the accuracy and responsiveness of system interaction but also emphasises the adaptability of users’ subjective cognitive load, emotional state, and behavioural habits, thereby expanding the application boundaries of multimodal interaction technology in ‘human–machine-context’ integrated cockpits.

3. Methods

The design process for an augmented reality (AR) game interaction design scheme based on situational awareness and virtual avatar design in an autonomous driving environment can be divided into the following three steps, as shown in Fig. 2.

(1) Interaction Mode Analysis Based on Situation Awareness Theory. The focus is on utilizing the situational awareness (SA) model of perception, comprehension, and projection for analysis. Human-machine interaction methods in L4 autonomous driving are integrated at this stage, where the system perceives the environment, understands user intentions, and predicts future situations.

(2) Scenario modelling based on the KANO model and user journey maps. Through the analysis of typical driving scenarios, scenario modelling is conducted, followed by element analysis. Additionally, user research is conducted using images and interviews to develop an AR solution, aimed at understanding user behavior and needs in different interactive scenarios.

(3) Experimental evaluation method based on eye tracking. During the solution design phase, based on the user behaviour analysis and priority ranking obtained in steps P1 and P2, key interaction scenarios are identified and corresponding augmented reality solutions are developed. Subsequently, a test environment is constructed using a simulator, and the effectiveness of the design solution is systematically evaluated using a combination of subjective questionnaire assessments and objective eye-tracking data. Based on this evaluation, iterative optimisation is carried out to improve the system’s interaction performance and user experience. This process integrates Situation Awareness,

human–machine interaction research, and user feedback evaluation to ensure the effectiveness and feasibility of the designed solution.

In summary, this study integrates SA theory with AR game design, and the main contributions at the methodological level are reflected in three aspects: theoretical integration, interaction modeling, and experimental validation. First, in terms of theoretical integration, SA theory is innovatively introduced from traditional aerospace, military, and human factors engineering fields into entertainment and safety fusion interaction design in L4-level autonomous driving smart cockpits. The “SA-driven semanticized game mechanism design” is proposed, which encodes the system prompts into visual elements with contextual semantics and perceptual salience through virtual characters such as “fire elves”, effectively guiding users to make prospective decisions and realizing the structured integration of theoretical mechanisms. Structured integration. Second, in terms of interaction modeling, an interaction scheme that maps the three levels of SA (perception, understanding, and prediction) to the visual cues of the game, the behavior of the virtual characters, and the real-time feedback mechanism is constructed. The cognitive function of AR cues in user environment modeling is clarified, and a systematic design framework of “cognition–interaction co-optimization” is proposed, which breaks through the previous interface-oriented single design mode. Finally, in terms of experimental validation, based on a driving simulator system, the proposed interaction mechanism is empirically analyzed in multiple dimensions by means of eye tracking, reaction time test and subjective trust assessment. The experimental results verify the effectiveness of the design scheme in enhancing users’ environment perception, decision-making efficiency and interaction trust, filling the research gap of insufficient support for situational awareness in the current intelligent cockpit, and forming a “cognitive assessment–behavioral performance–subjective load”. It forms a “cognitive assessment–behavioral performance–subjective load” closed-loop interaction system, which has good reusability and promotion value.

3.1. Interaction mode analysis based on situation awareness theory

To systematically support the interaction design of AR games in autonomous driving scenarios, this study introduces Endsley’s three-

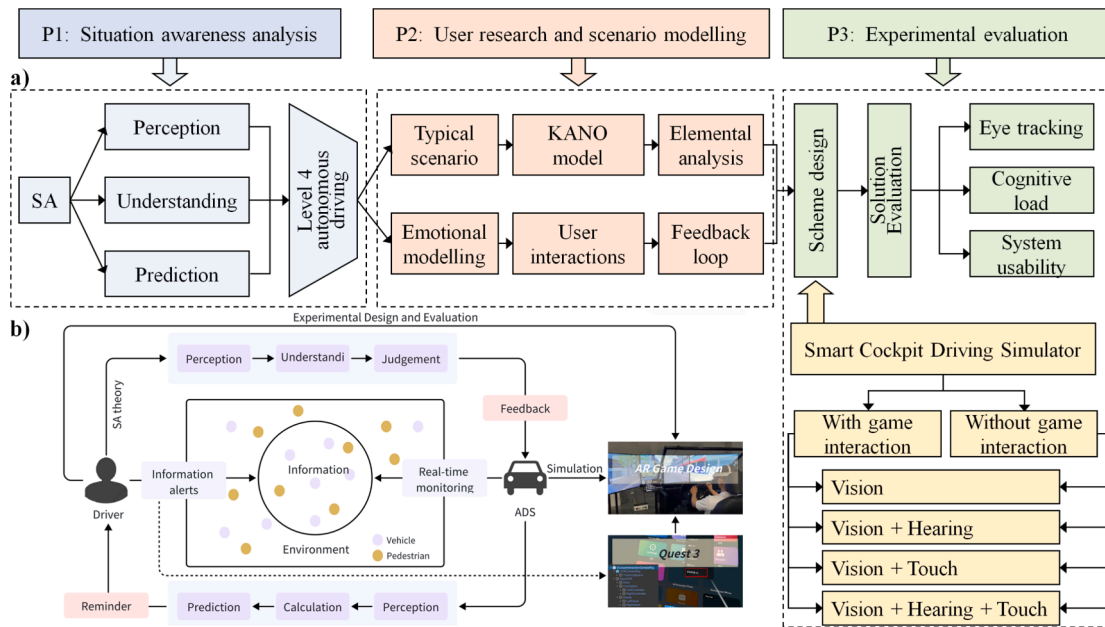


Fig. 2. Flowchart of the design and evaluation process for an augmented reality game based on situational awareness in an intelligent cockpit. a) Overall framework of the method, covering situational awareness modelling, user context modelling, and experimental evaluation modules. b) Specific implementation process of the proposed method in an intelligent cockpit simulation system, constructing a closed-loop driving simulation experiment path for human–vehicle–environment.

layer situation awareness (SA) model. It combines it with the ‘team SA–shared SA–fusion team and shared SA structure in multi-team systems to construct a multi-modal interaction analysis framework suitable for intelligent cockpits. This framework not only focuses on the real-time status recognition of individual drivers but also emphasises the dynamic coordination mechanism between individuals, virtual images, and system agents in the vehicle-mounted multi-agent system, as shown in Fig. 3 [28].

Level 1: Perception Layer. This layer emphasises the driver’s ability to perceive key elements in the current environment, including vision and spatial data collected by vehicle sensors, user cameras, and environmental perception systems (such as LiDAR and AR HUD). The system overlays core driving information (such as lane boundaries, pedestrians, and obstacles) as AR images in the driver’s field of view, ensuring the vision prominence and immediate communication of information, thereby enhancing the user’s engagement and alertness in the dynamic environment [29].

Level 2: Understanding Layer. At this level, the system integrates perception data to form a contextualised understanding of the user’s state and intentions. The virtual avatar acts as a cognitive agent, capable of sensing the user’s gaze, gestures, or voice input, inferring their current goals (such as entertainment or attention diversion), and dynamically adjusting the game content. The system also adjusts based on the user’s cognitive load, emotional state, and task engagement to ensure that game behaviour does not interfere with driving safety but instead assists in attention guidance at critical moments [30].

Level 3: Prediction Layer. This stage focuses on predicting the evolution of the situation. The system uses models such as trajectory prediction and risk assessment to proactively adjust game mechanics, alert users to potential risks, or provide suggestions and interactions through the avatar. For example, when the system predicts a potential hazard, the virtual avatar can guide the driver to redirect their attention, balancing interactive immersion with driving safety [31].

In multi-team systems, situational awareness construction is not limited to the individual dimension but also requires the introduction of structured mappings between team SA and shared SA. The first layer is Team SA, which emphasises individual independent cognitive processes around personal sub-goals, with limited information sharing; The second layer is Shared SA, reflecting information coordination and understanding among team members around common objectives; The third layer is the proposed integrated ‘Team and Shared SA’ (Ours), supporting collaboration mechanisms among multiple teams through shared common goals, thereby achieving stronger situational responsiveness and task coordination efficiency [32].

In summary, the interaction mode analysis framework constructed in this study, which combines the ‘three-layer SA model + multi-team collaborative structure,’ aims to achieve a closed-loop information flow throughout the ‘perception–understanding–prediction’ process in gamified AR interaction systems. It also ensures the system’s collaborative perception capabilities and safety response mechanisms in multi-role, multi-objective, and multi-task scenarios, providing a theoretical foundation and design support for immersive human–machine interaction in autonomous driving environments.

3.2. Scenario modelling based on the KANO model and user journey maps

3.2.1. User research based on user journey maps

This study explores user expectations of driver assistance features, in-vehicle entertainment, and gamified interactions in L4 autonomous driving scenarios through in-depth interviews and user research. The main goal of the study was to identify the scenarios in which drivers are most in need of enhanced road information perception in an automated driving environment and to improve their experience through gamification design. Through questionnaires and interviews, the study identified user experiences and needs in a computerized driving environment, with a particular focus on pain points and expectations related to entertainment interactions and safety information perception. Interview participants included 30 senior engineers with experience using self-driving cars, ranging in age from 22 to 33 years old and with driving experience from 1 to 10 years. Interview topics included their driving experience, in-vehicle entertainment behaviors, and expectations for self-driving assistance features and game design [33].

Through the above analysis, this paper constructs two representative user personas, as shown in Fig. 4. The first persona is the Young Experience Seeker, who craves innovative and personalized experiences. Long commutes or waiting in city traffic make them feel bored, and they seek to pass the time with novel vehicle features. These users have a strong interest in trying trendy, personalized functions and show a high level of enthusiasm for game-based interactions. The second persona is the Safety-Oriented User, who is more mature and cautious. They desire a high level of vehicle automation that can partially replace driving operations to alleviate the fatigue and stress of driving. They also have high expectations regarding the safety of intelligent systems and expect autonomous driving features to be practical and valuable. To further reveal users’ driving scenario preferences, interaction behavior patterns, encountered problems, as well as their expectations and preferences, a User Journey Map was created. This map helps identify more design opportunities and enables rational interaction design to improve the user experience.

The following four directions were distilled from the user journey maps to enhance gamified design to improve drivers’ perception of road information in autonomous driving scenarios. ①Achieve more intelligent, intuitive, and prominent driving assistance indicators through the game interface, ensuring that these indicators interact with the real-world environment, thereby enhancing the sense of technology and immersive experience; ②The game design should enable users to quickly grasp environmental information, effectively reducing their cognitive load while maintaining safety during the driving process; ③The system design should improve users’ understanding and trust in the autonomous driving system’s functions, thereby alleviating any potential anxiety during use; ④Through creative game interaction design, users can quickly respond to environmental information and clearly understand both current and future driving states. In summary, these research directions will guide the practice of future game system design, aiming to improve drivers’ situation awareness in autonomous driving vehicles while optimizing the overall driving experience.

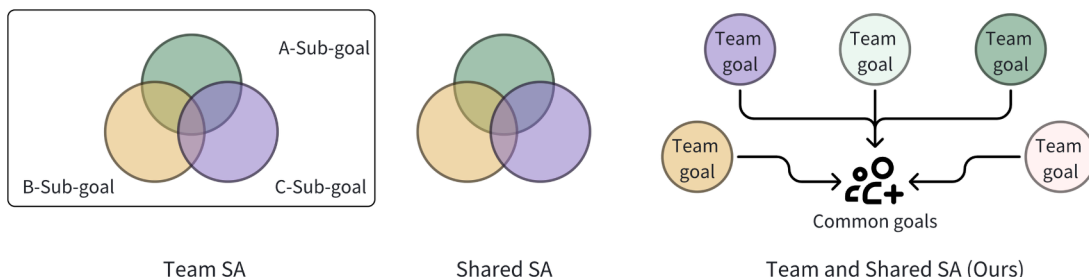
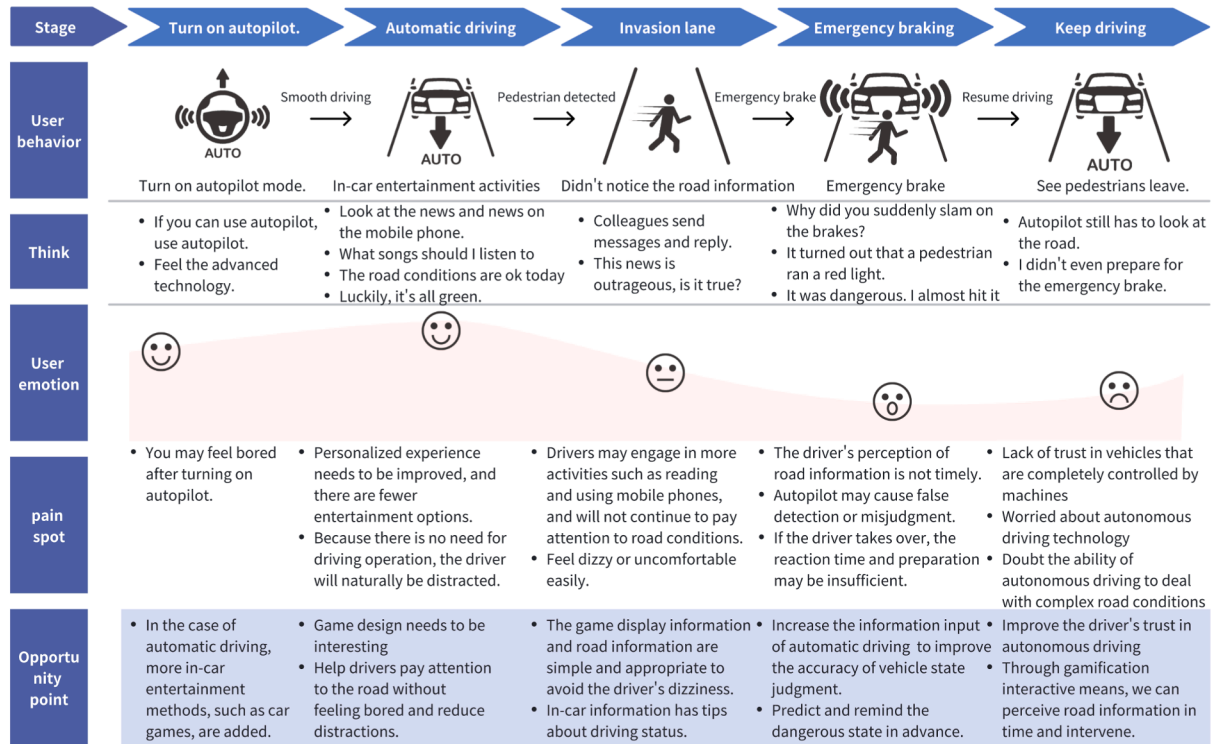
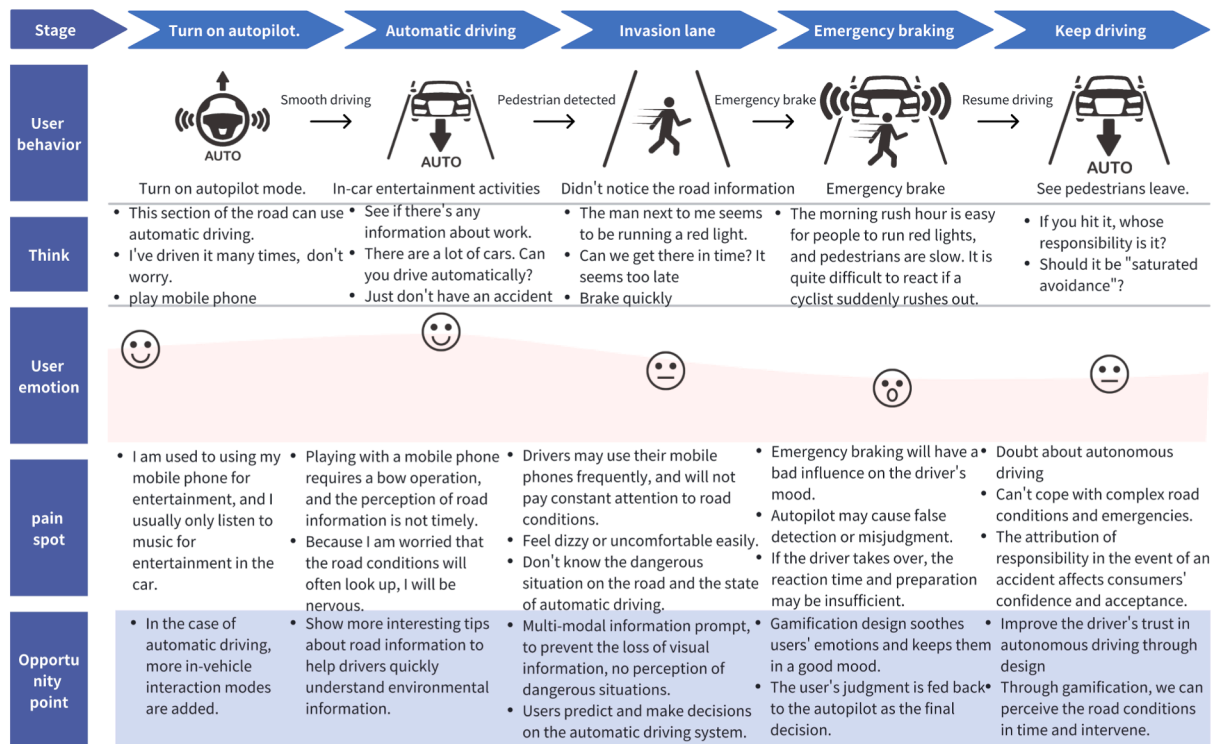


Fig. 3. L4-level autonomous driving behaviour analysis based on the fusion team and shared situational awareness theory.



a) Experience Seeker User Journey Map



b) Safety-Oriented User Journey Map

Fig. 4. User Persona Journey Maps.

3.2.2. User behaviour scenario modelling based on the KANO model

This study is based on the pain points and opportunity points identified in the user journey map. It employs the KANO Model to conduct a hierarchical analysis of user needs and constructs a demand priority matrix, using the degree of functionality and user satisfaction as dual axes, to reveal the non-linear relationship between product attributes and user satisfaction. The KANO Model categorises product attributes into five types: must-have, expected, delight, indifferent, and reverse: must-have attributes are the basic functions of a product, and their absence significantly reduces user satisfaction; expected attributes are positively correlated with satisfaction, meaning that the more complete the functionality, the higher the satisfaction; delight attributes exceed user expectations and significantly enhance satisfaction; indifferent attributes have limited impact on satisfaction; and reverse attributes may lead to user dissatisfaction. Combining questionnaire data, the study applied this model to the gamification design requirements analysis for Level 4 autonomous driving scenarios. Based on user satisfaction and functional availability, a two-dimensional coordinate axis was constructed to form a requirement priority matrix, as shown in Fig. 5, to identify and rank the importance and implementation urgency of various gamification scenarios, thereby providing a decision-making basis for subsequent interaction system design. In the specific screening process, driving safety was identified as the core evaluation criterion, requiring a comprehensive consideration of user demand intensity, scenario safety, and its frequency of occurrence in actual driving. By integrating the classification results of the KANO Model, safety analysis, and user experience feedback, the study systematically organised gamified scenarios that enhance drivers' road perception. In terms of geographical dimensions, urban and rural roads were assigned higher priority due to their higher complexity. In terms of environmental dimensions, although daytime and sunny conditions occur more frequently, rainy and nighttime driving scenarios were prioritised due to their higher urgency. In terms of emergency events, pedestrians suddenly entering the lane were rated as the most risky scenario, surpassing vehicles entering the lane, due to the greater unpredictability of

pedestrian behaviour. Based on the above analysis, the final prioritised test scenario is 'pedestrian intrusion at intersections on urban roads,' which will be validated under rainy night conditions to assess the game system's hazard warning effectiveness, while its universality will be tested under daytime sunny conditions. This strategic scenario selection approach aims to develop a multi-modal game interaction system that is both targeted and versatile, thereby enhancing drivers' situational awareness capabilities within autonomous vehicle cabins.

3.2.3. Typical scenarios for L4 autonomous driving games

Through user interviews and scenario analysis, it was found that most users trust the safety of autonomous driving technology, but some are concerned about the system's ability to handle unexpected situations and emergencies. Most users tend to listen to music or use their phones while riding, and they expect simple and easily interruptible mini-games to be added to the in-car experience. Users also hope that gamified interactions can incorporate road information to enhance immersion, without being overly complex or causing motion sickness. To improve drivers' information perception in complex environments, typical scenarios such as city roads, rural roads, rainy conditions, and emergencies were selected based on user feedback and rating mechanisms. In particular, during emergencies—such as when a pedestrian suddenly steps into the road or the car ahead makes an abrupt stop—users have an urgent need for assistance cues. These high-priority scenarios were used to design the multimodal interaction system [34].

In summary, this study proposes a gamified interaction design solution for L4 autonomous driving. The research results guide the design of multimodal interaction systems in autonomous driving cockpits to enhance drivers' situation awareness and user satisfaction. To further analyze the unique design elements and achievable interaction modes, a detailed interaction design framework was constructed. Based on the comprehensive analysis, a priority ranking of L4 autonomous driving scenarios was established. Table 1 outlines the key design elements of each typical scenario and the corresponding multimodal interaction modes, providing a reference for the design of multimodal game systems

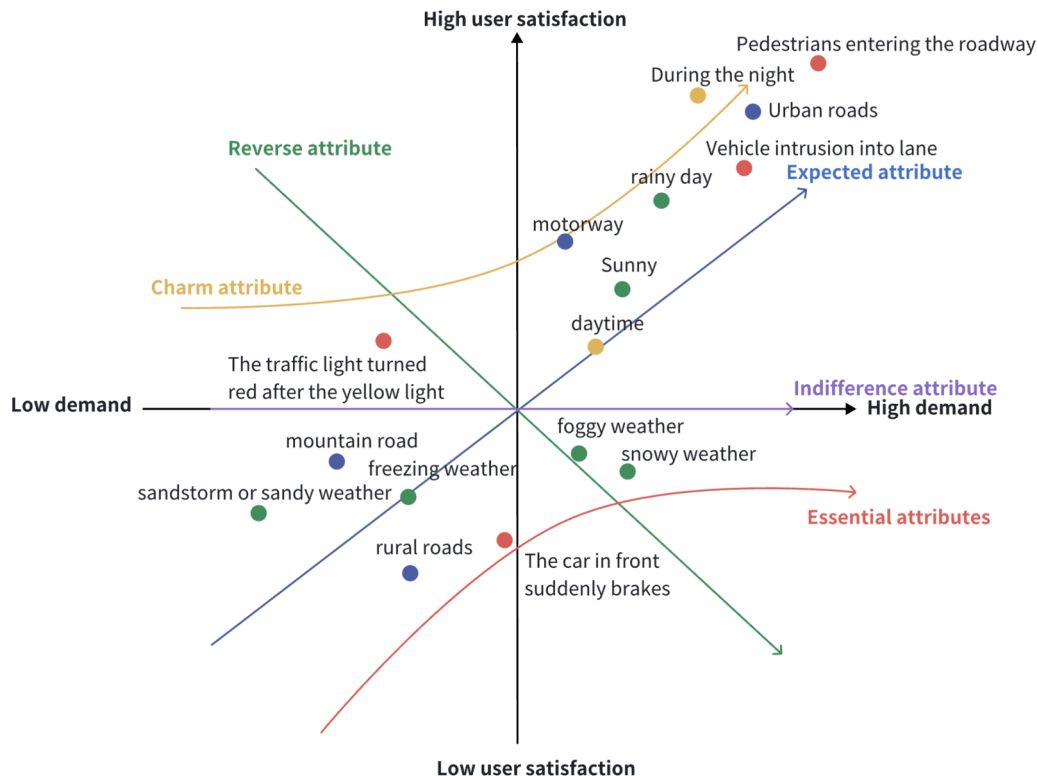


Fig. 5. Priority matrix ranking for Level 4 autonomous driving scenarios based on the KANO model.

Table 1
Priority ranking based on the KANO model in L4 autonomous driving scenarios.

Category	Scenario	Design elements	Potential interaction model
Geographic Area	Urban Roads	City buildings, roads, greenery, traffic lights, pedestrians, bicycles, traffic signs, etc	Vision
Weather	Rural Roads	Greenery, animals, etc	Vision
	Rainy Day	Visibility: High/Medium/Low/Rain	Vision, Hearing
Driving Time	Sunny Day	Sun, Temperature	Vision
	Night Driving	Brightness	Vision: Light changes
Emergency Situation	Daytime Driving	Brightness	Vision, Haptic: Temperature changes
	Pedestrian crossing into the lane at the intersection	Blind spot alert, pedestrian movement warning	Vision, Hearing, Haptic: Vibration
	A vehicle entering the lane from the side at the intersection	Blind spot alert, vehicle movement warning	Vision, Hearing, Haptic: Vibration

in intelligent cockpits [35].

3.3. Program design and evaluation

3.3.1. Scenario modeling for pedestrian avoidance at intersections

Based on the key user interaction points within the scenario, the pedestrian avoidance scenario at intersections is divided into four stages: normal driving, driver awareness of road conditions and detection of pedestrians, waiting for pedestrians to cross the intersection, and the vehicle continuing to drive. The details of each stage are outlined in Table 2.

Table 2
Scenario Modeling for Pedestrian Avoidance at Intersections.

Phase	Operating status	Related Description
Normal Navigation Driving		The vehicle is driving straight on an urban road at a constant speed of 40 km/h, with necessary driving information displayed to the passengers inside the vehicle through AR
Pedestrian Entering Lane at Intersection		As the vehicle prepares to pass through the intersection, a pedestrian suddenly enters from the right side, and the driver needs to perform an emergency brake to avoid the pedestrian
Waiting for Pedestrian to Pass		The car is waiting at the intersection for the pedestrian to pass
Continue Driving		The autonomous vehicle continues driving

3.3.2. Multimodal game task analysis

At the L4 autonomous driving stage, the vehicle's automation system assumes the primary driving tasks, while the driver shifts to a monitoring and decision-assistance role. Based on the team situation awareness model, the task design must ensure effective communication and collaboration between the driver and the autonomous driving system. The task design is as follows:

(1) Approaching the Intersection: The driver should focus on environmental monitoring and assess system performance through the in-car information system, ensuring the ability to intervene immediately if necessary. The autonomous driving system is responsible for vehicle control and providing key situational awareness information.

(2) Pedestrian Entering the Crosswalk: The driver needs to monitor the avoidance action in real-time, evaluating its appropriateness and sharing environmental information to assist in decision-making. The autonomous driving system autonomously decides and executes the avoidance procedure while continuously providing feedback on the operation status.

(3) Pedestrian Crosses and Normal Driving Resumes: The driver should verify whether the system restarts the journey as expected and maintain a monitoring role. The autonomous driving system ensures safe passage and continues driving, updating situational information for monitoring purposes. The multimodal game interaction design within the intelligent cockpit must display the following functional information:

(1) Basic Information: System status, settings, and time requirements.

(2) Safety-related Information: Navigation routes, destination, current location, driving direction, and road speed limits.

(3) Game Interaction Basic Information: Event prompts, feedback, and rewards.

(4) Road Information Elements: Alerts for road vehicles, pedestrians, and traffic light information. For the multimodal game interaction design solution in the intelligent cockpit, these functional information elements must be thoroughly considered and effectively displayed to meet the comprehensive interaction needs of the system [36].

Using Pokémon-style avatars, we aim to enhance user immersion and sustained attention through figurative, emotionally driven, and interactive feedback mechanisms. The multimodal task design is based on the following game psychology principles:

(1) Emotional Engagement and Character Anthropomorphism.

Visual anthropomorphism and Parasocial Interaction make it easier for users to emotionally connect with the virtual characters. More than 76 % of the users in the experiments reported that they “liked the characters and looked forward to subsequent interactions”, especially in the Pokémon style. More than 76 % of users in the experiment reported that they “liked the characters and looked forward to subsequent interactions”, especially when they received “character approval” after completing a task.

(2) Feedback Loops & Positive Reinforcement.

Whenever a user completes a critical task (e.g., timely reaction, successful obstacle avoidance), the character will perform an animated celebration/reward cue (e.g., “Great job!”, “Flash jump”). Whenever the user completes a critical task (e.g., timely reaction, successful obstacle avoidance), the character will celebrate with animation/reward prompts (e.g., “Great job!”, “Flash jump”), forming an instant positive feedback mechanism. The system builds a closed loop of task success-emotional response-reward prompts-next task motivation, which strengthens positive motivation.

(3) Adaptive Difficulty based on eye movement and task performance.

The system integrates an eye movement, gaze time and reaction time data analysis module, which automatically triggers the “Role Aid Mechanism” and the reward mechanism when distraction or a decrease in the success rate of a task is detected. The system automatically triggers the “role-assisted mechanism” and reward mechanism when

distraction or decreased success rate is detected, such as alerting the participant to increase the frequency of prompts, moderately decreasing the operation requirements, and alerting the user to the virtual character more explicitly (e.g., closer to the center of the visual line). In addition, the experiment set up a threshold warning, in which the character would automatically remind the user, “Be careful in front of you!” if the reaction time lasted > 1 s or the gaze dispersion exceeded the threshold, and simplify the task input process.

4. Experimental design

The three core research questions (RQs) of this study are as follows. To address the following questions, a 2 (with and without game interaction) $\times$ 4 (Vision, Vision + Hearing, Vision + Touch, Vision + Hearing + Touch) mixed factorial experimental to verify the effectiveness of the proposed method.

RQ1: Can multimodal AR games, supported by situational awareness theory, effectively enhance drivers’ perception of road and environment information?

RQ2: Can gamified interaction design in smart cockpits effectively reduce drivers’ reaction time in unexpected events?

RQ3: Does multimodal game interaction help to improve drivers’ trust and ride enjoyment of the autonomous driving system?

These three research questions jointly validate the effectiveness of the proposed SA-based AR gamified interaction framework in improving information perception, response efficiency, and user experience. RQ 1 is validated in 5.1 Task Performance Analysis, RQ 2 is validated in 5.2 Physiological Data Analysis, and RQ 3 is validated in 5.3 Subjective Evaluation.

4.1. Scheme design

4.1.1. Element extraction and design

This paper systematically models the interactive functions of multimodal games based on L4 autonomous driving scenarios and provides a detailed classification of the information categories involved. Building on this foundation, the paper further focuses on the core design elements of multimodal games, aiming to optimise the visibility, interactivity, and guidance of the game interface through a three-tiered element extraction mechanism (‘L1: Perception—L2: Understanding—L3: Prediction’), thereby enhancing users’ environmental perception, cognitive response, and entertainment experience in autonomous driving environments.

Specifically, game elements are divided into three layers: the perception layer (environmental information display), the understanding layer (interaction behaviour and feedback), and the prediction layer (guidance and prediction), corresponding to users’ intuitive acquisition of environmental information, immediate response to interaction behaviour, and proactive understanding of system status and future scenarios. The Perception layer emphasizes real-time awareness of the external environment. Its design elements include rainy and nighttime vision effects, urban street scene backgrounds, traffic sign recognition, and ambient sound effects. These features help users intuitively understand the current driving conditions. The Understanding layer focuses on the interaction process between user input and system feedback. It encompasses virtual character animations, interactive container design, emotional feedback mechanisms, and dynamic sound effects, thereby establishing a closed-loop interaction between game events and user actions. The Prediction layer guides users in making anticipatory judgments about system operation and task flow. Design elements such as scoreboards, prompt windows, and control buttons enhance the user’s control over game progress and driving status. Together, these three layers aim to construct a complete interaction chain from sensory input to cognitive decision-making. For detailed classification and analysis, see Table 3.

For example, in this system, the SA of L1 is embodied in the instantaneous linkage of virtual game characters (e.g., fire sprite, water

Table 3

Element Hierarchy, Attributes, and Detailed Descriptions.

Layer	Attributes	Key Elements
L1	Intuitive perception of the interface and environment	Rainy night effects: raindrops, puddles Urban background: buildings, streetlights Traffic elements: signs, lights, pedestrians, vehicles Ambient sounds: rain, vehicles
L2	User interaction and system response	Fairies: animated characters with expressions Warning signs: speed limits, obstacles Capture tools: nets, containers Sound effects: captures, alerts Fairy animations: flying, escaping
L3	Experience evaluation and game flow control	Scoreboard: progress and score Tip box: instructions, warnings Buttons: start, pause, settings

sprite) with road environment information. When the system recognizes a specific scene state (e.g., pedestrian crossing ahead, water on the road, etc.), it will visualize the key environmental elements (e.g., emergency braking is prompted by a flame symbol, or the risk of water pooling is prompted by a water droplet) through AR visual cues to ensure that the user receives a clear, intuitive, first-level environmental input in the AR interface. In L2, the system comprehensively judges the user’s current intention or cognitive state through the user’s interaction behavior (e.g., reaction time, gaze point) and emotional state (subjective emotion scale or facial expression capture), and dynamically adjusts the interface complexity and cueing method. For example, when the system detects that the user has repeatedly missed prompts or delayed responses, the virtual character will actively simplify the content of the information prompts, reduce the cognitive load, or add voice feedback for guidance, to realize the system’s “understanding” of the user’s state and make strategic adaptations. At the L3 level, the system combines the road model with behavioral prediction logic (e.g., there is a potentially dangerous point in front of the vehicle), and the virtual character appears in advance and gives obvious “pre-action” prompts. For example, the Fire Wizard appears in the AR interface before the user observes pedestrians, and is accompanied by sound or vibration cues to direct the user’s attention from entertainment content to road risks, thus providing a “cognitive preview” of future events before the user is aware of the problem. This mechanism of perception-understanding-prediction makes the AR game not just a decorative entertainment component, but a dynamic interaction module that enhances the driver’s situational awareness.

Additionally, from the perspective of vision and interactive expression, game elements can be further categorised into four major types: background elements (e.g., road environments, weather effects), interactive elements (e.g., clickable objects, feedback animations), user interface elements (e.g., prompt panels, score bars), and sound effects (e.g., ambient sounds, interactive sound effects). These categories support the coordination and sustainability of multi-modal systems in terms of vision consistency, interactive rhythm, and cognitive guidance. For details, see Table 4 [37].

The multimodal game prompts designed in this study adopt cute animal-like sprite characters. In terms of gameplay, a “Pokémon-style” interaction model is mimicked, allowing users to capture sprites within the game. The sprite character designs are shown in Table 5, and the modeling was done using Maya 2018 software. Through this series of design and modeling efforts, the goal is to create a multimodal game interaction experience that not only provides effective situation awareness support but also offers fun and interactivity. The mood board includes the game prompt information at the top and the game feedback information at the bottom [38].

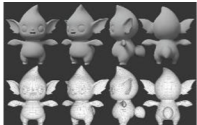
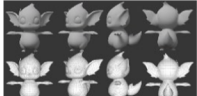
Table 4
Classification of different game elements and detailed display.

Element Category	Element	Details
Background	Urban Background	Buildings, streets, streetlights, billboards, etc
	Weather Effects	Raindrops, puddles, rain mist, slippery roads, raindrops hitting the windows, etc
	Traffic Elements	Traffic signs, traffic lights, pedestrians crossing streets, other vehicles, etc
Game Interaction	Sprites	The main characters of the game, featuring different animations and expressions
	Scoreboard	Displays the driver's score and game progress
User Interface	Start/Pause Button	Allows the driver to start or pause the game
	Settings Button	Allows the driver to adjust game settings
Sound & Music	Interaction Sound Effects	Clicking sounds, capture sounds, warning sound effects, etc

4.1.2. System information layout design

In this study, the information displayed in the multimodal game interaction solution was carefully constructed based on the typical driving scenarios described earlier. The information content is divided into four core sections: system information, driving information, game information, and road information. The system information section covers fundamental data such as the current status of the autonomous driving mode, time, and weather conditions. Driving information includes key indicators such as vehicle speed, navigation route, vehicle location, guidance lines, and maps. Game information focuses on in-game event prompts, corresponding user feedback, and reward mechanisms. Road information provides detailed real-time data on road conditions, including pedestrians, vehicle flow, and traffic light indicators. See Fig. 6 for details [39].

Table 5
Design proposals and descriptions for fairy characters with different elements.

Attribute	Time	Three-View Model	Mood board	Explanation
Fire	Emergency Braking by the Car			When pedestrians or vehicles enter the driveway and need to stop urgently, burning flames will appear on the game interface.
				When the driver observes the dangerous road environment alarm (the fire elf appears to indicate) and will implement emergency braking, he will provide interactive feedback immediately.
Water	Passing through a puddle on a rainy day			When the vehicle is about to pass through a puddle on a rainy day, a Water Sprite appears on the game interface.
				When drivers observe the warning of road puddles (indicated by the appearance of water sprite), they need to provide interactive feedback immediately.
Wood	Passing through a rural road			When the car is driving on a rural road and greenery appears by the roadside, a Wood Sprite appears on the game interface.
				When the driver observes the environmental prompt—indicated by the appearance of the Wood Sprite—they must immediately provide interactive feedback. After the feedback is received.

Through a systematic evaluation and ranking process, this study further selected the two scenarios with the highest comprehensive scores in each category. For these selected scenarios, we conducted an in-depth analysis of their unique design elements and achievable interaction patterns to establish a detailed interaction design framework. Subsequently, based on the comprehensive analysis results, as detailed in Table 6, the key design elements of each typical scenario and their corresponding multimodal interaction patterns are systematically elaborated, aiming to guide the design of multimodal gaming systems in intelligent cockpits. Through this research methodology, not only can the genuine needs of users in autonomous driving scenarios be understood, but also tailored gamified interaction experiences that meet user expectations can be designed, thereby enhancing drivers' situational awareness levels in autonomous driving environments.

4.2. Program development

Considering the characteristics of Augmented Reality (AR) vision presentation, which involves the blending and overlaying of virtual information with the real world, this study carefully controlled the use of colors when designing the AR vision scheme for the intelligent cockpit. To ensure that passengers can accurately recognize interface information during driving and minimize vision distractions caused by an excessive number of interface elements, the design scheme adopted a blue-themed color strategy as the main interface for passenger interaction with the AR system. Common components serve as the foundational elements for constructing the AR vision interface. These common components are divided into four categories: system information, driving information, game information, and road information. The system information design elements include icons for system status, time, weather conditions, and the activation or deactivation of vibration and sound settings. Driving information includes icons for navigation route maps,

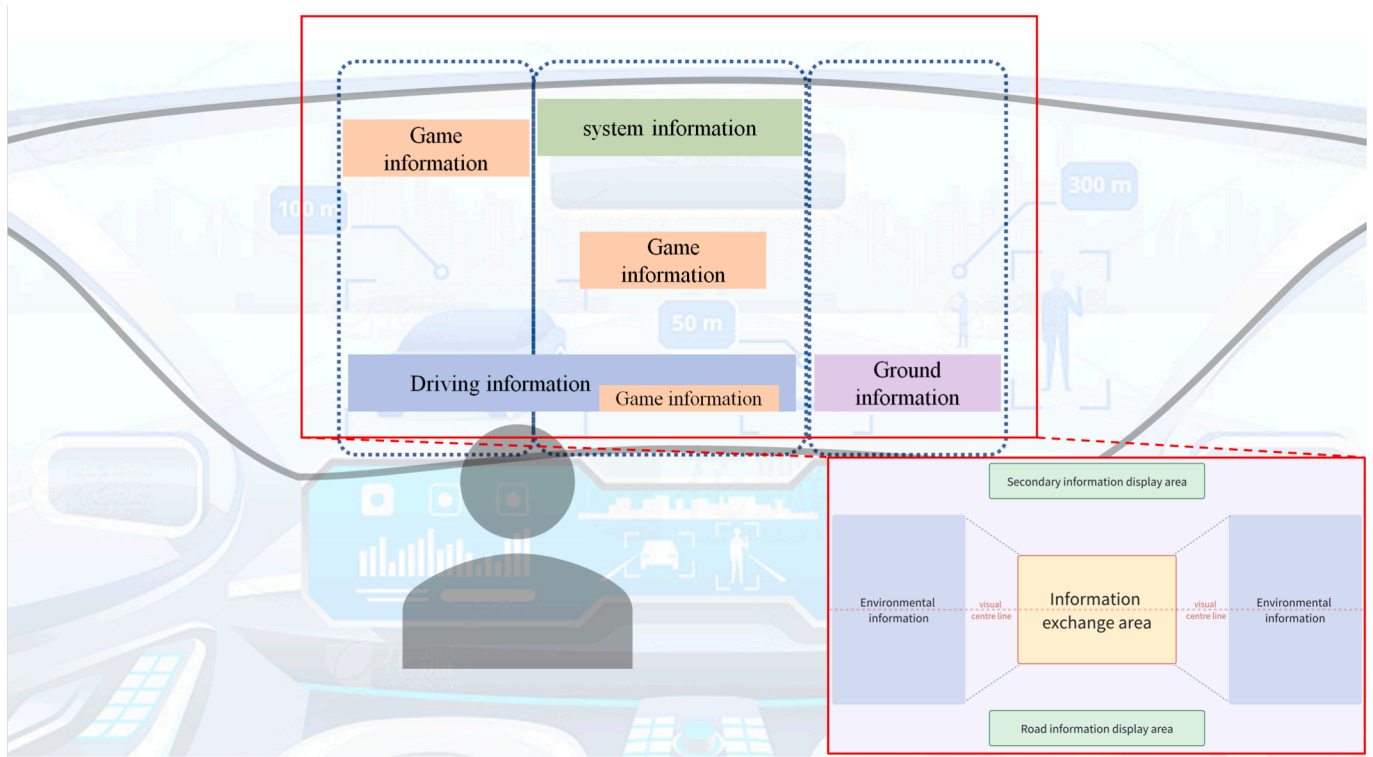


Fig. 6. System Information Layout.

Table 6
Categorization of L4 Autonomous Driving Scenarios, Design Elements, and Modalities.

Category	Typical Scenario	Representative Design Elements	Interaction Modalities
System Information	Pedestrian intrusion at intersection	Blind spot warnings, pedestrian trajectory prediction	Vision, Hearing, Touch
	Side vehicle intrusion at intersection	Side vehicle detection, collision trend prediction	Vision, Hearing, Touch
	System operating status	Takeover prompts, power/energy status, network connectivity	Vision, Hearing
Driving Information	Rainy weather (light/medium/heavy)	Raindrops, slippery road, blurred visibility	Vision, Hearing, Touch
	Sunny weather	Sunlight intensity, blue sky, glare	Vision
	Nighttime driving	Low lighting, light reflections, headlight illumination	Vision
	Daytime driving	Natural lighting, temperature changes	Vision, Touch
Game Information	Multimodal task prompts	Game objectives, progress bars, score indicators	Vision, Hearing, Touch
	Virtual character interaction	Virtual NPCs, emotional expressions, voice dialogues	Vision, Hearing, Touch
	User input behaviors	Gestures, button clicks, gaze-based interaction	Vision, Touch, Hearing
Road Information	Urban roads	City buildings, road structure, greenery, traffic lights, pedestrians, bicycles, signs	Vision
	Rural roads	Rural architecture, natural vegetation, animals, simplified road signs	Vision

autonomous driving cruise indicators, and speed limits. A virtual steering wheel displaying the direction of autonomous driving was specifically designed to enhance the user's immersive experience with the interface. The game information components introduce an experience point progress bar and a monetary reward system, based on optimization suggestions from participants in pre-experiments. This design aims to increase interactivity and playability by visualizing reaction times as measurable values, while also motivating users to improve their response speed. Road information icons include road navigation guidance lines, blind spot vehicle alerts, traffic light status, and pedestrian warnings. The detailed layout is shown in Fig. 7.

In the AR system design of this study, the SA model not only serves as a theoretical reference framework but also directly guides the key decisions on the timing of information presentation, content prioritization, and multimodal selection in the interface design, which are reflected as follows:

(1) Timing of Interface Presentation – From the “Perception” Strategy Based on Level 1 (Perception) of the SA model, we emphasize the pre-event cueing mechanism in the design. Perception” strategy. Based on Level 1 (Perception) of the SA model, we emphasize the pre-event

prompting mechanism in our design. For example, when the system detects a potential pedestrian crossing event 20 m ahead, the “fire sprite” in the system will appear in advance, even if the pedestrian has not yet entered the field of view. Through visual prompts, sound reinforcement, and vibration co-presentation, it guides the driver to carry out early warning reaction, thus realizing the activation of perceptual preoccupation and avoiding the traditional HUD problem of “information lag”.

(2) Content Prioritization – Cognitive Load Control Based on “Understanding”. In Level 2 of the SA model, we take “Understanding” as a reference. In Level 2 of the SA model, we regard “Understanding” as the system's recognition and matching of the user's current attention and workload. A task prioritization control mechanism is introduced in the design: in the area where the user's eye movements are concentrated, safety-related cues (e.g., pedestrians, red lights, and speed) are presented in preference to secondary game points or animations. At the same time, according to the user's reaction time and task completion, unnecessary prompts are dynamically reduced to reduce visual cognitive interference and optimize comprehension efficiency.

(3) Multi-modality Choice – Risk Level Grading Mechanism



Fig. 7. General Component Design Illustration.

Embodying “Prediction”. Under the guidance of Level 3 (Prediction) of the SA model, we choose different modality combinations for prompts according to the urgency of the event. For example, for low-risk events, only visual cues are used; for medium-risk events, such as “slippery road in the rain”, sound effects are added; and for high-risk events (e.g., pedestrian crossing), a combination of three modes, namely, visual, sound and vibration, is used to ensure that the user’s attention is focused and a timely response is provided. Through the modal combination of the graded response system, the user’s cognitive simulation and behavioral preparation for future risks are enhanced.

In summary, our design implements the SA three-layer model as follows: presenting key content in advance, dynamically adjusting the information density, and providing risk-graded feedback through modal combinations, forming an adaptive AR interaction mechanism centered on the user’s state and environmental changes.

To combine real-world scenarios with game interaction, the AR game was developed within an AR environment. The software and hardware used during the development process include the following: Unity 2022—used for programming and exporting the AR game; Meta Quest Developer Hub—used to install the AR game onto the Quest 3 platform; Quest 3—the AR game runtime platform; USB 3.0 data cable—used for debugging the AR game, connecting the Quest 3 hardware to the Unity

platform. Based on the research, the AR game environment was built. Due to the unique characteristics of AR game design, the background is made transparent to blend UI game elements, interactive icons, and real-world road background information, achieving an AR interactive gaming effect. In this game, grasp interaction and laser interaction were utilized. Once the game is successfully installed, it can be accessed by clicking on “Unknown Sources” in the library to locate the installed AR game. The final effect is shown in Fig. 8.

The experimental environment was set up in a car studio, utilizing a driving simulation system independently developed by the laboratory. During the experiment, participants used a lever to activate the autonomous driving function. The driving simulator was equipped with a computer connected to a three-channel fusion panoramic screen, creating an experimental environment that closely resembles real driving experiences for the participants. In Quest 3, participants used right-hand gesture interactions to capture fire sprites.

4.3. Preliminary experiment

To validate the effectiveness of multimodal gamified interaction design in enhancing drivers’ situational awareness (SA) levels within intelligent cockpits, this study conducted a pre-experiment before the

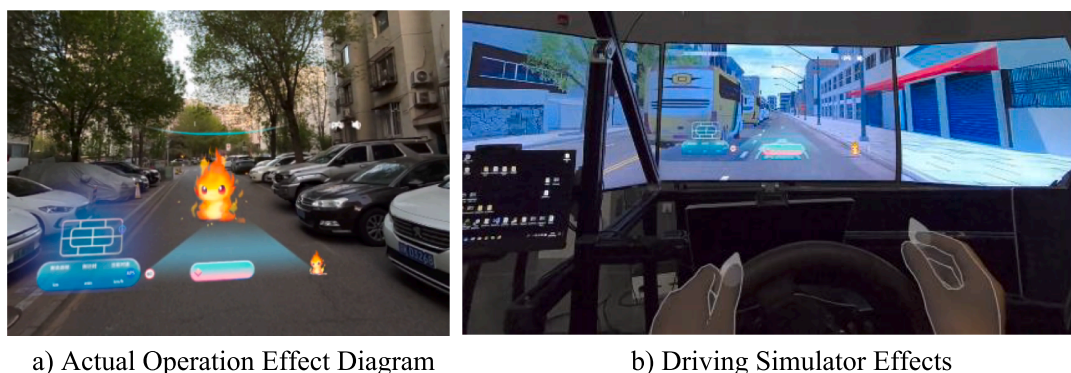


Fig. 8. Software Development Effect Diagram.

formal experiment. The pre-experiment systematically assessed the advantages of gamified mechanisms over traditional non-gamified interaction schemes in terms of perception efficiency and user experience, and further compared the differences between single-modal and multi-modal feedback.

This pre-experiment included an interaction group under traditional manual driving conditions as a baseline reference to compare the specific improvement in SA levels across different interaction design schemes in L4 autonomous driving scenarios, thereby enhancing the causal inference capability regarding the impact of gamified interaction on SA levels.

Based on the priority test scenarios determined in Section 3.2.2, the User behaviour scenario modelling is based on the KANO model. The pre-experiment simulates a ‘nighttime rainy urban intersection’ scenario, as shown in Fig. 9. This scenario is derived from prior user research and features low visibility and high alertness requirements, making it suitable for situational awareness sensitivity assessment. The experimental platform is equipped with a visual-auditory-tactile feedback system, including a triple display, steering wheel vibration motor, and Bluetooth feedback module, to achieve tactile simulation, as detailed in Fig. 9a.

A total of eight interaction schemes were set up for this preliminary experiment, divided into two categories: a traditional non-game group and a game interaction group. Each category contains four modal combinations: Vision (V), Vision + Hearing (V + H), Vision + Touch (V + T), and Vision + Hearing + Touch (V + H + T) (Fig. 9b). Among these, the ‘Water Spirit’ and ‘Wood Spirit’ events were only used for interactive teaching and were not included in the formal data testing.

The experimental route was set as a two-way circular roadway (Fig. 9c), with the vehicle travelling at a constant speed of 40 km/h. Four pedestrian event trigger points were set at six symmetrical intersections. Whenever a ‘pedestrian’ or ‘fire spirit’ event is triggered, the participant must immediately press the response button. Reaction time is calculated as the time difference between the event trigger and button operation. Each interaction scheme is tested once, with the average reaction time recorded across four trigger points. After the experiment, participants complete the SA questionnaire.

For haptic feedback, small vibration motors were installed in the steering wheel, which vibrated immediately upon event triggering; participants could stop the vibration via a Bluetooth button. Due to issues such as difficulty in controlling vibration intensity and excessive interference observed in multiple trials, it is planned to replace the vibration scheme with the NorthGrip game controller, which features precise vibration control functionality, in the formal experiment to enhance interaction stability.

The experimental results showed that compared to non-game interaction conditions, the introduction of game mechanisms significantly shortened the participants’ reaction time, with the ‘V + H’ multimodal combination yielding the optimal effect. Additionally, some participants reported that the current interface had excessive information density in the night rain scene, posing a risk of visual overload. They suggested further simplifying the interface design and increasing interaction richness (e.g., through a points reward mechanism) to enhance participation motivation. Furthermore, some participants pointed out that while the night rain scenario is highly challenging, it lacks representativeness and generalisability. They suggested using a ‘daytime sunny’ condition in formal experiments to enhance the generalisability of the results. After comprehensive consideration, the formal experiment scenario will be adjusted accordingly to obtain more representative experimental data.

During the pre-experimental phase before the formal experiment, the research team identified and addressed potential discomfort issues associated with wearing AR headsets, a problem that has been widely reported in previous VR/AR studies. To enhance the acceptability of the experimental experience, reduce confounding variables, and ensure the validity of experimental data, this study optimised equipment selection and experimental procedures based on feedback from the pre-experiment, specifically including:

(1) Controlling the duration of each experiment: Each round of the experiment was limited to 15 min, with mandatory rest periods between multiple rounds to alleviate eye strain and physical fatigue;

(2) Selecting lightweight equipment: The Meta Quest 3 AR headset was used, which features a lighter weight and an adjustable headband mechanism, effectively reducing neck and head pressure;

(3) Adjusting seat and viewing distance configurations: Standardising the calibration of driving simulation seats and display screen heights to maintain a horizontal line of sight, reducing visual strain, and optimising human-machine posture adaptability;

(4) Collecting subjective comfort feedback: After the experiment, administering a comfort assessment questionnaire covering dimensions such as visual fatigue, pressure sensation, and motion sickness. Results showed that over 85 % of participants reported ‘no significant discomfort’ or ‘acceptable,’ indicating the system has good ergonomic adaptability.

Additionally, to further enhance experimental stability and participant experience, researchers maintained real-time observation and established an exit mechanism during formal experiments to ensure experiments could be promptly terminated if severe discomfort occurred.

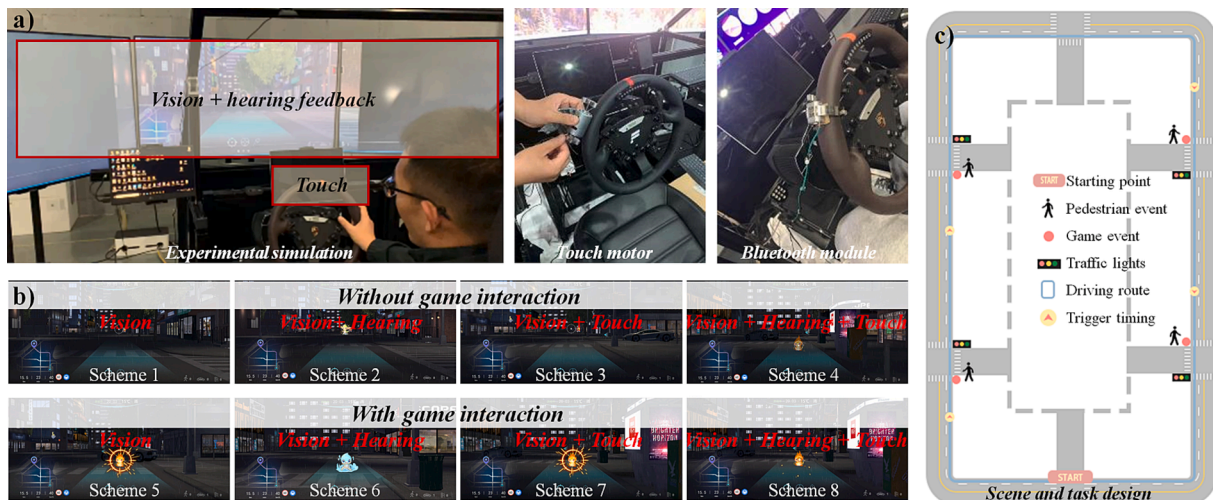


Fig. 9. Preliminary experiment scenario design and vision plan, including experiment scenario setup and software and hardware environment configuration.

4.4. Final experiment

4.4.1. Experimental tasks and procedures

To achieve the experimental objectives, eight comparative experiments were designed, divided into two major categories: one category involved no game interaction in autonomous driving mode, while the other involved game interaction, as detailed in Fig. 10. In each scenario, the effects of Vision, Hearing, and Touch modalities were explored as control variables. In the formal experiments, two different interaction schemes were set up to assess the impact on drivers' cognitive abilities and perception of road information under autonomous driving conditions.

In the pre-experimental phase, this study conducted a preliminary exploration of the temporal presentation of multimodal interaction cues, setting up two conditions of synchronized presentation (concurrent presentation) and staggered presentation for comparative testing. The results show that the synchronized presentation strategy performs better in enhancing users' perception speed and response accuracy to dangerous situations. Therefore, in the formal experiments, we adopted a parallel design in which Vision, Hearing, and Touch cues were activated at the same time to minimize the information processing delay and improve the response efficiency in the driving task.

It should be noted that the formal experiments did not systematically analyze the timing effects of interleaved presentation, mainly for the consideration of driving safety and avoiding cognitive confusion, to ensure that the multi-channel cues can work together and reach the user's perceptual system simultaneously. Future research can further design experimental conditions with different modal combinations and presentation timings (e.g., visual advancement, auditory lag, etc.) to further explore the effects of inter-modal synergistic strategies on perceptual load, task performance, and user experience.

The without game interaction scheme aimed to establish a baseline for drivers' cognitive levels under autonomous driving conditions. In this scheme, drivers were required to complete the following tasks during driving: ① When passing through an intersection, if the driver noticed a pedestrian crossing from the side, they must immediately press the button. ② When passing through an intersection, the system may issue a Hearing alert when a pedestrian is crossing, and the driver must press the button after identifying the potential danger. ③ Similarly,

when a pedestrian crosses the road, the system may emit a vibration alert, and the driver must press the button after identifying the danger. ④ When a pedestrian crosses the road, the system may emit both a Hearing and vibration alert, and the driver must press the button after identifying the potential danger.

This game interaction scheme aims to assess the driver's ability to perceive road information under gamified interaction conditions. In this scheme, the driver must complete the following tasks: ① When passing through an intersection, the system may use vision effects to remind the driver to pay attention to pedestrians crossing the road. The driver must respond quickly and press the button to receive in-game currency as a reward. If the reaction time exceeds 1 s, a certain amount of reward will be deducted. ② Similar to the above, but warnings will be provided simultaneously through visual effects and audio prompts. ③ Similar to the above, but warnings will be provided simultaneously through vision effects and vibration feedback. ④ Similar to the above, but warnings will be provided simultaneously through vision effects, audio prompts, and vibration feedback. Drivers must press the button immediately after identifying the hazard to receive a reward; otherwise, if the reaction time exceeds 1 s, a certain amount of virtual currency will be deducted as a penalty.

To effectively control for order effects (such as learning effects or fatigue effects) that may arise from within-subject designs and interfere with experimental results, this study employed a Latin square design to balance the presentation order of the eight interaction conditions. Specifically, within each participant group, the order of condition combinations experienced by each participant was unique, ensuring that different modality combinations (e.g., V, V + H, V + T, and V + H + T) appeared in different positions across participants. This approach balanced any systematic biases that might arise from the order of presentation of interaction conditions.

Through this counterbalancing randomisation mechanism, the experimental results more accurately reflect the impact of different interaction conditions on participants' cognitive performance and subjective load, significantly enhancing the internal validity of the experiment.

4.4.2. Reward mechanism design

To quantify users' response efficiency and reward feedback under

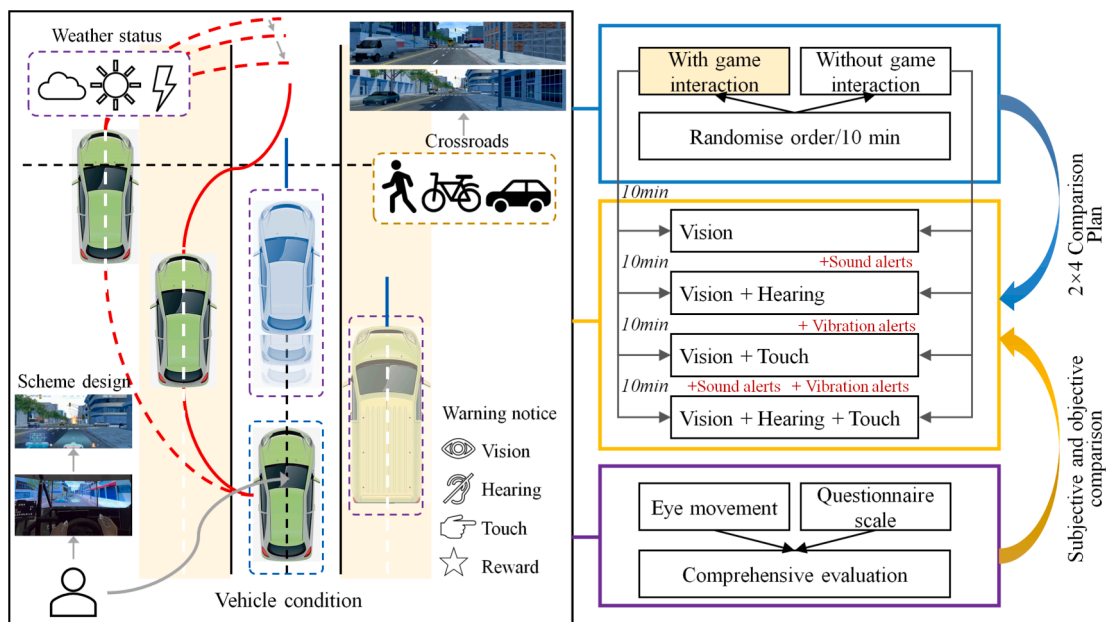


Fig. 10. Formal experimental flowchart, including two types of interaction schemes and four modal combinations, constituting a total of eight experimental conditions. All interaction conditions are presented in a balanced order among different subjects through a Latin square design to control sequence effect interference.

different modal prompts, this paper designs two conversion functions based on reaction time. These are used to generate experience scores (S) and monetary rewards (M), respectively, to construct instant feedback and challenging mechanisms. Both are constructed based on piecewise linear functions and are formulated concerning driver reaction limits and dangerous response thresholds (0.1 s to 1 s). The relationship between the reaction time t and the experience score s is defined as a piecewise function.

$$S(t) = \begin{cases} 100, & t \leq 0.1 \\ -111.11 \cdot t + 111.11, & 0.1 < t \leq 1 \\ 0, & t > 1 \end{cases} \quad (1)$$

Among them, when the driver reacts extremely quickly (≤ 0.1 s), they receive a full score of 100 points; between 0.1 s and 1 s, the experience score decreases linearly; when the reaction time exceeds 1 s, the score is reset to zero.

The monetary reward function is designed as another linear piecewise function, defined as shown in $M(t)$.

$$M(t) = \begin{cases} 100, & t \leq 0.1 \\ -125 \cdot t + 112.5, & 0.1 < t \leq 0.9 \\ -125 \cdot t + 112.5, & t > 0.9 \end{cases} \quad (2)$$

When the reaction time exceeds 0.9 s, the monetary reward is negative (penalty); between 0.1 s and 0.9 s, it decreases linearly; those with extremely short reaction times receive the maximum positive reward.

In this experiment, the system calculates the reaction time t by recording the difference between the event trigger time and the user's button response time, and uses it as an input variable in the behavioural feedback function to generate empirical values and virtual currency rewards in real time, thereby constructing a 'quick response-instant incentive' game feedback mechanism. To enhance interaction intensity and task challenge, this study sets '1 s' as the critical behavioural threshold: when the reaction time is less than or equal to 1 s, the system provides positive incentives (high scores and rewards); when the reaction time exceeds 1 s, users may receive low scores or have currency deducted, simulating the negative feedback associated with potential risks.

According to research in human factors engineering and traffic safety, the average reaction time in sudden driving situations is approximately 0.7–1.0 s, with specific values varying depending on factors such as situation complexity, prompt modality, and driving experience. For example, in a study on the effects of sensory interference on driving reaction times, Reference [40] noted that in typical emergency scenarios without high levels of interference, drivers' average reaction times were concentrated in the 0.75–0.95-second range, with the 95 % confidence upper limit not exceeding 1.1 s. Therefore, this paper adopts '1 s' as the threshold in the game feedback mechanism, which not only has a reference basis in psychological and human factors research but also balances the experiment's interpretability, game tension, and system feasibility.

It is important to emphasise that this study does not treat '1 s' as an absolute critical threshold for driving safety, but rather as a controllable threshold within the behavioural feedback mechanism to distinguish between high and low performance. This is aimed at enhancing the immersiveness and engagement of the experiment while maintaining relative consistency with the conclusions of human factors research.

In addition to compensating participants for their time and cooperation in the experiment, this study provided each participant with a monetary reward of 50 yuan after the experiment concluded. This amount was approved by the ethics committee during the pre-experiment phase and is considered moderate compensation, insufficient to significantly influence participants' behaviour or motivation, thereby ensuring the authenticity and scientific validity of the experimental data. All compensation was distributed in a single payment after the experiment concluded and was not directly linked to experimental

performance, aligning with standard practices for incentive control in behavioural research.

Therefore, this study suggests that '1 s' can be used as a judgment criterion for the effect threshold of modal combination before the distraction exceeds the help: when the multimodal cueing makes the driver's reaction time exceed 1 s, it is possible to change from 'cognitive synergy' to 'perceptual load' overload. When the multimodal cue makes the driver's reaction time exceed 1 s, it may change from 'cognitive synergy' to 'perceptual load', which suggests that the system needs to optimize the modal combination strategy or adjust the cueing density and tempo, to avoid the counter-effects of distraction.

4.4.3. Final experimental design scenario

Based on feedback from the pilot experiment, the formal experimental virtual scenario was set as a daytime city road scene. The resolution of the experimental system was kept consistent with the resolution of the driving simulator. The simulator's display consisted of three screens, each with a resolution of 3840x2160 pixels, stitched together horizontally, so the experimental system's resolution was set to 11520x2160 pixels. The formal experimental scenario involved a city intersection where pedestrians needed to be avoided. The experimental design featured an autonomous driving navigation route that continued straight at the intersection, but suddenly, a pedestrian would appear on the right, crossing the street against the red light, requiring the vehicle to decelerate promptly. In terms of speed, the main vehicle drove at a constant speed of 40 km/h throughout. During the experiment, there were six intersections, four of which involved pedestrians suddenly entering the road. The experiment focused on measuring the driver's reaction time in detecting pedestrians entering the roadway and the need to engage emergency braking in this hazardous situation. To keep the environment similar to typical city road conditions, the traffic situation was kept smooth, with no significant variations from normal traffic.

The virtual scene was built using Unity 2022, applying the Fantastic City Generator plugin to generate the basic city road model and simulation traffic system, making elements such as vehicles and traffic lights in the scene more realistic. Based on vision color design standards, the design of common components, and feedback from the pilot experiment, the AR experimental solution used 3D modeling to recreate the daytime city road and programmed the typical scenario of pedestrian avoidance at intersections. Realistic game visions and interactions were used to demonstrate the multimodal game AR solution within the intelligent cockpit. See Fig. 11 for details.

To control for potential bias and learning effects in the eight experimental conditions, the following measures were taken in this study:

(1) Condition Order Randomization and Latin Square Balanced Design. All experimental conditions were presented in a pseudo-randomized order (pseudo-randomization), and the order of the experiments for the 30 subjects was controlled by grouping them based on the principle of Latin Square (Latin Square) balancing. It was ensured that each condition appeared in a balanced position across subjects to avoid learning, fatigue, or expectation bias due to a specific order.

(2) Task length control and intermediate rest design. Considering the long total experimental time of the eight conditions, each experimental task was limited to about 10 min, interspersed with 5–10 min of active rest passages, during which the system guided the user to carry out light activities (e.g., head movement) to reduce the accumulation of cognitive load and somatosensory fatigue, and thus to reduce the interference of the decline of performance in the later stages of the experiment.

(3) Tutorial Training and Proficiency Warm-up Phase. Uniform practice tasks (not counted in the results) were set up, and pre-experiments were conducted before the formal experiments to ensure that all participants had a basic familiarity with the task logic and AR mechanism before entering the main experiment, thus minimizing the impact of learning curve differences on the results.

These strategies effectively mitigated the potential interference



a) Passing Through the Intersection.



b) Triggered Event



c) Interaction Feedback



d) Autonomous Driving Cruise

Fig. 11. Overall Vision Scheme for Multimodal Game.

caused by order dependence, expectation bias, and repeated exposure during the experiment, and enhanced the reliability and interpretive power of the data.

4.5. Evaluation indicators and parameterization

In this study, experiments were conducted using a driving simulator with the core objective of investigating the impact of applying multimodal game interaction design in specific scenarios on drivers' situation awareness levels and subjective experience in the context of autonomous driving. By comparing non-gamified interactions with gamified interactions, as well as single-modal, dual-modal, and multimodal interaction methods in an autonomous driving environment, the study aims to evaluate the effectiveness of multimodal game interaction solutions in enhancing drivers' perception of road information. Additionally, the study seeks to verify the feasibility of game interaction design within the intelligent cockpit. At the same time, the experiment aims to collect user feedback on the game solution to provide data support for future design optimizations. In conjunction with the cognitive indicators and evaluation methods mentioned earlier, this section will evaluate the design solution from three key dimensions: subjective scale data, task

performance, and physiological indicators. These three dimensions cover 10 specific data metrics, as outlined in Table 7.

To ensure quality and validity control of the experimental data, we used a full within-subjects design in which all 30 participants completed the driving task in 8 experimental conditions (2 interaction types $\times$ 4 pattern configurations). To minimize order effects, we used a Latin variance design to randomize the order in which participants were presented with the conditions. Statistical analyses were performed using a mixed-effects ANOVA model with interaction type and mode as fixed effects and subject ID as a random effect to account for individual differences. The model was applied to Physiological Indicators (e.g., N1 ~ N2), Subjective Cognition (e.g., N3 ~ N9), and Task Performance (e.g., N10) to ensure robustness of inferences. Data analysis was done in SPSS 26 and JASP software.

4.6. Experimental setup

4.6.1. Experimental equipment

Since eye-tracking data needed to be collected and the Quest 3 headset does not include an eye-tracking module, and because the pilot experiment revealed that external vibration motors significantly

Table 7
Different design scheme evaluation dimensions and corresponding indicators.

Evaluation Dimension	Aspect	Specific Indicator	Number	References
Physiological Indicators	Gaze Focus	Eye-tracking heat map	N1	[41]
	Gaze Path	Eye-tracking path map	N2	
Subjective Cognition	Workload	NASA-TLX scale	N3	[42]
	Situation Awareness	SAGAT	N4	
	Acceptance	Acceptance scale	N5	[43]
	Trust	Trust scale	N6	
	Self-Assessment of Emotion	SAM scale	N7	[45]
	System Usability	SUS system usability scale	N8	
	Subjective Evaluation	In-depth interviews	N9	[41]
Task Performance	Task Response Time	User reaction time	N10	

interfered with participants, the formal experiment utilized a Beitong game controller with built-in vibration motors for game interaction. After the experimental events were triggered, participants recorded the time by pressing the X button on the controller. The hardware equipment used included a Beitong game controller (with built-in vibration motor), a driving seat, a simulation environment display screen, a car simulation framework model, and an eye tracker.

4.6.2. Experimental participants

This experiment employed a multi-factor analysis of variance (ANOVA) to assess the effects of prompt modality (within-subjects factor) and the presence or absence of game interaction (between-subjects factor) on driving behaviour, attention allocation, and reaction time. Based on pre-experimental data, in autonomous driving scenarios, game interaction and multi-modal prompts have a moderate to large effect on driving performance. Therefore, statistical power analysis was conducted using GPower 3.1 software, as shown in Fig. 12.

GPower is a software tool for statistical power analysis that helps researchers predict the required sample size before experimental design. The effect size f was set to 0.25 (moderate effect), the significance level α to 0.05, and the statistical power to 0.8. The number of experimental conditions was 8 (2×4 combination), i.e., presence or absence of game interaction (2 levels) $\times$ prompt modality (4 levels: vision, vision + Hearing, vision + Touch, vision + Hearing + Touch). Under this design, the degrees of freedom for the game interaction factor are 1 (2–1), the degrees of freedom for the prompt modality factor are 3 (4–1), and the degrees of freedom for the two-factor interaction term are $1 \times 3 = 3$. According to the calculation results, the total sample size required to achieve the set statistical power is 179, meaning that each group requires approximately 23 participants ($179 \div 8 \approx 22.375$).

The final experiment recruited a total of 30 participants (13 males and 17 females), aged between 19 and 25 years, with an average age of 22.35 years. All participants had normal or corrected vision, held valid driving licences, and had driving experience ranging from 1 to 5 years. All participants completed a 15-minute training session on the simulated driving system and interface adaptation before the formal experiment to ensure they possessed the necessary operational proficiency and interface understanding required to perform the tasks. All participants signed informed consent forms, the experiment underwent ethical review, and appropriate compensation was provided to the participants.

It should be noted that, given the pioneering and exploratory nature of this study, which introduces AR game interaction design methods into autonomous driving scenarios, the experimental sample primarily focused on young drivers and users with some gaming experience. This sample selection somewhat limits the generalisability of the research

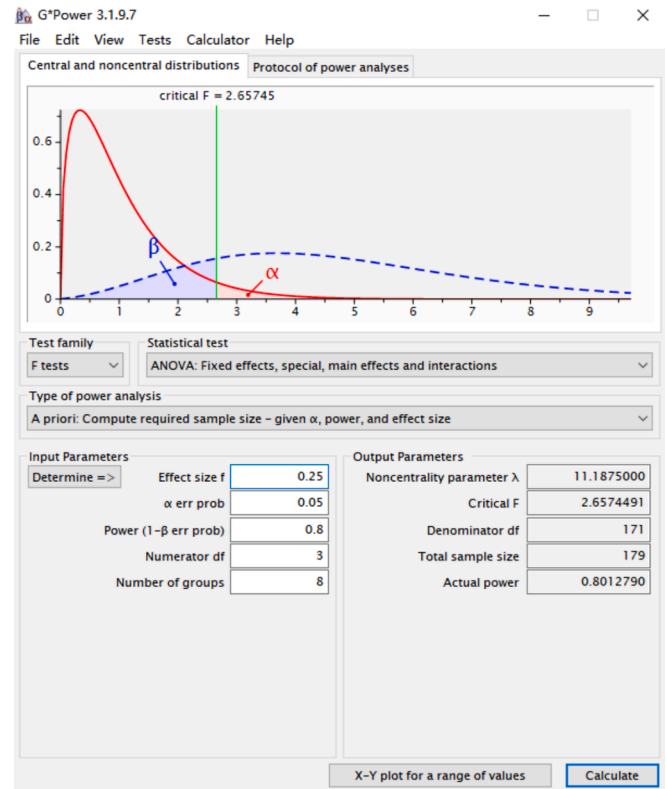


Fig. 12. Sample size calculation graph (using GPower 3.1, setting medium effect size $f = 0.25$, significance level $\alpha = 0.05$, and statistical power of 0.8).

results, particularly in terms of extrapolation to older users and non-gamers. Considering that future users of autonomous driving systems will encompass a broader range of ages and usage backgrounds, including those who are sceptical about autonomous driving technology, future research will further expand the sample composition to include middle-aged and elderly drivers, non-gaming users, and individuals with varying degrees of acceptance of emerging technologies, thereby enhancing the representativeness and practical application value of the research findings.

5. Experimental results

5.1. Task performance analysis (RQ1)

Task performance includes task completion rate, task response time, monetary rewards, and experience points earned from tasks, as detailed in Table 8. The task completion rate refers to the participant completing the button press action within 1 s after triggering the pedestrian event.

Table 8
Average response time, score, and reward performance under various combinations of prompt modalities and game interaction conditions.

Scheme	Modality Factor	Time/s	Average score	Reward
Without game interaction	Vision	0.8968	11.47	0.40
	Vision + Hearing	0.6900	34.44	26.65
	Vision + Touch	0.5836	46.27	66.20
	Vision + Hearing + Touch	0.5250	52.78	113.08
With game interaction	Vision	0.6961	33.77	25.49
	Vision + Hearing	0.5418	50.91	70.27
	Vision + Touch	0.4996	55.59	120.31
	Vision + Hearing + Touch	0.4521	60.87	176.29

During this timeframe, the autonomous vehicle is considered to be in a state where it will not collide with the pedestrian. If the button press is not completed within 1 s, the task is deemed a failure. In this experiment, 29 participants were involved, with 25 participants able to complete the button operation within the specified time. However, 4 participants were unable to complete the button operation within the specified time under the non-game interaction scheme.

Task reaction time refers to the time from the pedestrian event trigger to the user pressing the button. Statistical analysis of reaction times for all participants in both schemes revealed that the Shapiro-Wilk test indicated non-normal distribution characteristics ($p = 0.031 < 0.05$), while non-parametric tests showed significant differences in reaction times between the two schemes ($p = 0.003 > 0.05$). The average reaction time for the game interaction scheme was overall better than that for the non-game interaction scheme, as shown in Fig. 13.

The multimodal approach combining vision, hearing, and touch modalities resulted in the shortest reaction times under both conditions. Next was the vision-Touch modality, followed by the vision-Hearing modality, which was slightly slower than the vision-Touch modality. The single vision modality had the longest reaction time. The average reaction time for the non-game interaction scheme was 0.67 s, with a variance of 0.02 and a standard deviation of 0.14. The average reaction time for the game interaction scheme was 0.55 s, with a variance of 0.008 and a standard deviation of 0.09. For details, please refer to Table 8.

To further validate the effectiveness of the multi-modal game interaction scheme in autonomous driving scenarios, this paper compares and analyses the reaction times of drivers under two schemes: 'no game interaction' and 'game interaction.' Statistical indicators include mean, variance, and standard deviation, with results shown in Table 9. The data results show that the game interaction scheme outperforms the non-game interaction scheme in terms of average reaction time (0.5474 s) and variability (standard deviation of 0.09150), with the non-game interaction scheme having an average reaction time of 0.6738 s and a standard deviation of 0.14166. This indicates that gamified design can effectively reduce drivers' reaction time to hazardous events and improve efficiency in handling sudden situations.

Further analysis reveals that the average reaction time for the Touch modality in the multimodal design is significantly better than that for the Hearing modality, indicating that Touch stimuli can more quickly trigger drivers' reaction behaviours. Therefore, their application in emergency alerts should be prioritised in actual design. Additionally, based on user interviews and system log analysis, the integration of gamification and multimodal design can enhance users' sensitivity to

Table 9

Average reaction time, variance, and standard deviation for the two schemes.

Indicator	Without game interaction	With game interaction
Mean	0.6738	0.5474
Variance	0.02007	0.00837
Standard deviation	0.14166	0.09150

environmental information, thereby improving the overall safety and effectiveness of the interaction system.

Given that this study employs a within-subject design, in which the same participant completes tasks under multiple interaction conditions, the data exhibit repeated measurements and a hierarchical nested structure. To further enhance the accuracy of statistical inference and the robustness of model fitting, this paper incorporates a linear mixed-effects model (LMM) for an in-depth analysis of reaction time data.

In the model, the two interaction schemes and four modality combinations were set as fixed effects, while participant ID was set as a random effect to control for individual differences in reaction times.

The results showed that the interaction scheme had a significant main effect on reaction time ($F = 9.23$, $p = 0.004$). The multimodal combination also had a significant main effect ($F = 6.87$, $p < 0.01$). There was a significant interaction between the two ($F = 4.12$, $p = 0.029$). The model's marginal $R^2 = 0.46$, conditional $R^2 = 0.73$, indicating that fixed and random factors together explain most of the variance in the data.

The model results are consistent with the conclusions of the nonparametric test, further enhancing the robustness and explanatory power of the study in handling repeated measurement data. As shown in Fig. 13 and Table 8, the three-mode fusion interaction achieved the shortest average response time in both scenarios, indicating that multimodal integration helps to improve user perception efficiency. The research results show that the multimodal game mechanism that introduces virtual roles and visually enhanced cues significantly enhances the users' recognition efficiency and response sensitivity to the surrounding environmental information (e.g., pedestrians, obstacles, weather conditions, etc.), which effectively supports the value of the three-phase model of situational awareness for application in interaction design.

5.2. Physiological data analysis (RQ2)

5.2.1. Eye movement path analysis and heat map comparison

The Tobii Pro Lab software was used to process and analyze the

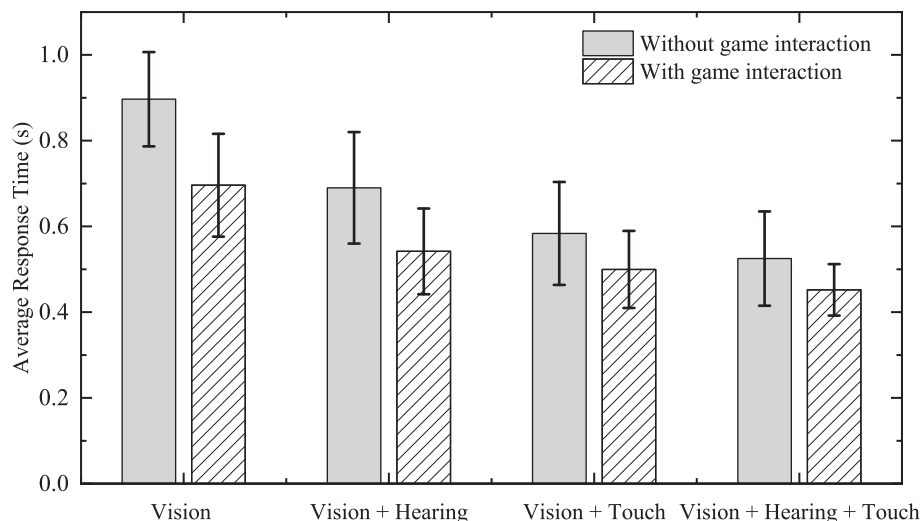


Fig. 13. Average reaction time for different interaction design schemes.

collected eye-tracking footage. Video segments showing changes in the participants' field of view during the tasks were extracted, and areas of interest (AOIs) were marked. After matching, gaze path maps and heat maps were generated. In the gaze path maps, the size of the circles represents the duration of fixation at a particular location, and the numbers within the circles represent the sequence of fixations, indicating the participant's gaze-following pattern. In the heat maps, the color intensity of the hotspot areas reflects the duration of fixation, ranging from red (longer) to green (shorter), and indicates the distribution of participants' attention and their focus on specific information.

As shown in Table 10, because participants needed to monitor the right side for potential pedestrian intrusions, their vision focus was predominantly distributed toward the right side of the display. In the non-game interaction scheme, users had to independently search for pedestrians, assess whether they would enter the lane, and determine if a dangerous situation requiring emergency braking might occur. This led to more scattered gaze paths and a more dispersed distribution of attention. In contrast, in the game interaction scheme, participants' attention was mainly focused on the area in front of them where the game effects appeared, tending to observe potential hazards at a central and horizontal level in their field of Vision. Interviews revealed that participants felt that, in the game scheme, the appearance of the Fire Sprite effect signaled that a pedestrian about to enter the lane would also appear to the right side. As a result, they no longer needed to monitor

Table 10

Analysis of user gaze heatmaps and gaze trajectory diagrams based on eye tracking data under different modalities and interaction schemes.

Scheme	Modality Factor	Indicated Heat Map	Indicated Path Map
Without game interaction	Vision		
	Vision + Hearing		
	Vision + Touch		
	Vision + Hearing + Touch		
With game interaction	Vision		
	Vision + Hearing		
	Vision + Touch		
	Vision + Hearing + Touch		

other pedestrians and vehicles on the road as closely, making their observation more purposeful. The timing and position of the Fire Sprite helped participants know when to check the side areas, improving observation efficiency and effectively alleviating their anxiety. By comparing the reaction time measurements under different interaction scenarios, it can be seen that the multimodal interaction mechanism (Vision + Hearing + Touch) can prompt drivers to react more quickly in the face of emergencies, and the average reaction time is shortened by about 0.12 s compared with that of the non-gamification scenario, which verifies the significant role of game interaction design in enhancing the efficiency of emergency response.

5.2.2. Average pupil diameter comparison

Pupil diameter is associated with emotions, motivation, and attitudes. The average pupil diameter during task completion can reflect the psychological state of the subject, such as emotional state and tension level. The greater the psychological effort, the larger the pupil dilation. Based on the average pupil diameter data of the subject's left and right eyes collected by the eye tracker during the task period, the average pupil diameter data of each scheme in the two scenarios was calculated, as shown in Fig. 14. Average pupil diameter of each design scheme. The average pupil diameter of the vision single-modality design in the non-game interaction scheme and game interaction scheme was 3.97 and 3.77, respectively. The vision + Hearing bimodal design had average pupil diameter values of 3.68 and 3.59 in the two schemes, respectively. The vision + Touch bimodal design had average pupil diameter values of 3.72 and 3.68 in the two schemes, respectively. The vision + Hearing + Touch multimodal design had average pupil diameter values of 3.58 and 3.51 in the two schemes, respectively.

The average pupil diameter in the game interaction condition was smaller than that in the non-game interaction condition, indicating that game interaction can help participants maintain a calm emotional state and alleviate tension. The average pupil diameter in the multimodal design condition was smaller than that in the bimodal design condition, suggesting that multimodal design can also effectively alleviate tension. In the bimodal design, the average pupil diameter for the vision + Hearing modality was smaller than that for the vision + Touch modality. Through user interviews, it was found that Touch modality vibration feedback designs are relatively rare, and from a sensory perspective, they are more stimulating than Hearing modalities, causing sudden alertness and tension, but this improves significantly with habituation. Additionally, considering the pleasure scores mentioned earlier, the game interaction scheme yields higher pleasure scores than the non-game interaction scheme. Therefore, from both subjective and objective perspectives, the game interaction scheme can effectively alleviate drivers' tension when encountering road hazards under autonomous driving conditions and enhance their sense of pleasure.

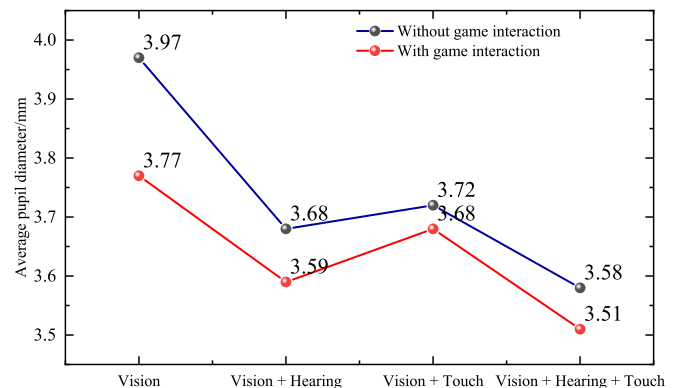


Fig. 14. Comparison of average pupil diameter among different design schemes.

5.3. Subjective evaluation (RQ3)

5.3.1. Comparative analysis of user acceptance and emotional responses under different interaction schemes

After completing the experimental tasks for both the ‘non-game interaction’ and ‘game interaction’ prompt schemes, all participants completed a subjective acceptance questionnaire based on the Technology Acceptance Model (TAM). The scale consists of 12 items, covering three dimensions: perceived usefulness, perceived ease of use, and satisfaction. Each dimension includes four questions, and the scale uses a 7-point rating system (1 = strongly disagree, 7 = strongly agree). Before conducting statistical analysis, the score data for the three dimensions were subjected to a Kolmogorov–Smirnov (K-S) normality test. The results indicated that all data followed a normal distribution ($p = 0.073 > 0.05$), satisfying the prerequisites for subsequent parametric tests. An independent samples *t*-test was then used to compare the perceived usefulness, perceived usability, and satisfaction data between the ‘no game interaction’ and ‘game interaction’ experimental conditions. The results showed that perceived usefulness ($p = 0.875$), perceived usability ($p = 0.630$), and satisfaction ($p > 0.05$) did not reach statistical significance, indicating no significant differences between the two interaction methods in terms of subjective acceptance. Although the statistical tests did not reveal significant differences, a comparison of the mean values shows that the ‘game interaction’ condition scored higher across all three dimensions: perceived usefulness scored $M = 5.75$ ($SD = 1.52$), perceived usability scored $M = 5.50$ ($SD = 1.21$), and satisfaction scored $M = 4.50$ ($SD = 1.28$); while the average scores for the ‘no game interaction’ scheme were $M = 5.00$ ($SD = 1.43$), $M = 3.80$ ($SD = 1.11$) and $M = 3.75$ ($SD = 1.25$), respectively. As shown in Fig. 15a, the overall trend indicates that game interaction may have certain advantages in improving the user experience. According to the interview results, the gamified interaction scheme significantly improved the prominence and recognisability of the target by designing road pedestrian prompts as ‘fire spirits,’ thereby reducing the visual search burden. Additionally, the fire spirit’s capture effect provided users with immediate feedback upon button activation, further enhancing the immersion and perceptibility of the interaction. These design factors may be the primary reason for its slightly higher acceptance scores compared to the non-game interaction scheme.

To further analyse the impact of different interaction methods on users’ subjective emotional states, this study introduced the Self-Assessment Manikin (SAM) method. Participants’ emotional states under the two schemes were evaluated across three dimensions:

pleasure, arousal, and dominance, using a 9-point scale (1 = very low, 9 = very high).

First, a K-S test was conducted on the data for the three emotional dimensions. The results indicated that the data did not follow a normal distribution ($p < 0.05$), so a non-parametric test (Wilcoxon signed-rank test) was used to compare the two schemes. The results showed that there was a significant difference in Pleasurability between the two interaction schemes ($p = 0.005 < 0.01$), with the game interaction scheme ($M = 7.4$) significantly higher than the non-game interaction scheme ($M = 6.3$); arousal levels also showed significant differences between the two conditions ($p = 0.007 < 0.01$), with the game interaction scheme scoring significantly higher ($M = 7.6$) than the non-game interaction scheme ($M = 6.7$); There was no significant difference in advantage level ($p = 0.213 > 0.05$), with both schemes exhibiting moderate to high levels (game-based interaction $M = 6.8$, non-game-based interaction $M = 6.1$).

As shown in Fig. 15b, the results indicate that the gamified prompt mechanism can effectively improve users’ emotional arousal and pleasure in autonomous driving scenarios, helping to alleviate negative experiences such as indifference and dullness; while the sense of superiority is mainly influenced by factors such as task structure and control permissions, and no significant difference was observed.

5.3.2. Comparison of subjective workload in interactive tasks based on NASA-TLX

To quantitatively assess differences in cognitive load among users in autonomous driving scenarios under different prompting methods, this study introduces the NASA Task Load Index (NASA-TLX) as a subjective workload measurement tool. The NASA-TLX comprises six evaluation dimensions: mental demands, physical demands, time pressure, task performance, self-efficacy, and frustration. Each dimension is scored on a scale of 0–10, with higher scores indicating stronger subjective workload perceptions in that dimension. In addition to scoring, participants were required to use a paired comparison method to weigh the importance of the six dimensions. By comparing each of the 15 dimension pairs, participants selected the one they considered more important in the task, with the number of votes serving as the weighting basis for that dimension. Combining weights and scores, the final weighted average was calculated as the overall workload index to measure differences in subjective workload perception under different interaction conditions.

The experiment conducted a Shapiro-Wilk normality test on the collected total workload scores, which showed that the data followed a

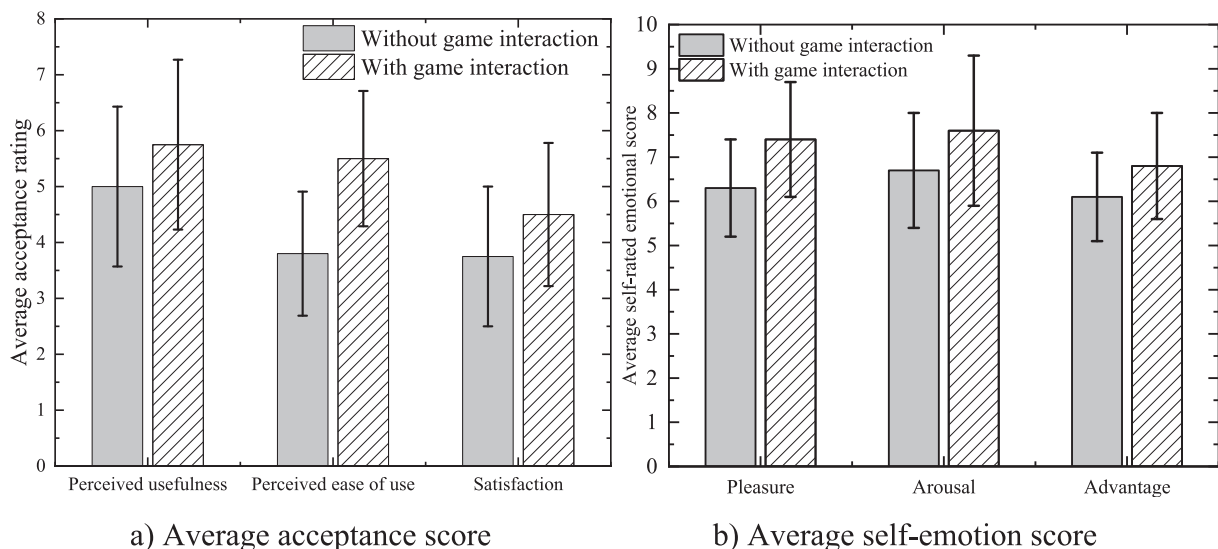


Fig. 15. Analysis of user acceptance and emotional response results under different interaction schemes.

normal distribution ($p = 0.676 > 0.05$), meeting the prerequisites for parametric testing. Subsequently, a one-way analysis of variance (ANOVA) was conducted to compare the workload differences between the 'non-game interaction' and 'game interaction' conditions. The results satisfied the homogeneity of variance test (Levene's test, $p = 0.827 > 0.05$), and the differences between the two groups were statistically significant ($F = 3.426$, $p = 0.039 < 0.05$), as shown in Fig. 16.

Further analysis of the six-dimensional scores revealed that under the 'with game interaction' condition, users' subjective load scores on all dimensions were lower than those under 'non-game interaction,' which was particularly significant in terms of psychological needs, time pressure, and frustration. This indicates that game mechanisms have certain advantages in alleviating users' cognitive load. Based on interview data, most participants reported that the sudden appearance of the 'Fire Spirit' event enhanced their vigilance toward potential hazards (such as pedestrians ahead), and the clear interaction semantics reduced the ongoing monitoring pressure in complex scenarios. Additionally, some participants interpreted the appearance of the 'Fire Spirit' as a prompt for the vehicle to automatically decelerate, thereby reducing anxiety and unease during sudden events. Overall, participants experienced a significant reduction in the cognitive resources and psychological tension invested during task completion, resulting in lower perceived subjective load across all dimensions.

In summary, experimental data and user feedback jointly validate that, compared to traditional non-game-based prompting schemes, interaction designs incorporating game mechanics can effectively reduce user workload in autonomous driving environments while enhancing interaction experiences, particularly demonstrating significant advantages in high-load scenarios such as low visibility (e.g., nighttime rain).

5.3.3. Assessment of the impact of interaction design based on the SAGAT and SUS scales on user situational awareness and trust

(1) Situation Awareness Analysis.

This study used the Situation Awareness Global Assessment Technique (SAGAT) to measure participants' situation awareness levels. Given the short duration of the experimental tasks, the SAGAT questionnaire was designed to be completed immediately after each group of tasks to avoid task interruptions and attention distractions. The questionnaire content covers the three levels of situational awareness: L1 Perception, L2 Understanding, and L3 Prediction, requiring participants to recall and answer questions related to the experimental scenario. Each question was scored as 1 point for a correct answer and 0 points for an incorrect answer, with the final score reflecting the participant's situational awareness level. A Shapiro-Wilk normality test was conducted on the total situational awareness scores under the two

interaction conditions. The results indicated that the data did not follow a normal distribution ($p = 0.000 < 0.05$), so non-parametric tests were selected. The Wilcoxon rank-sum test was used to compare the 'non-game interaction' and 'game interaction' conditions, and the results indicated a significant difference between the two ($p = 0.013 < 0.05$), as shown in Table 11. In terms of average scores, the overall situational awareness score for the game interaction scheme ($M = 88.9$) was significantly higher than that for the non-game interaction scheme ($M = 78.8$), indicating that gamified prompt mechanisms can help improve drivers' overall situational awareness of the road environment in autonomous driving scenarios.

Further analysis of the dimensional performance of the three levels of situational awareness involves pairwise comparisons between L1 (perception), L2 (understanding), and L3 (prediction). The results show that there are significant differences between the two schemes at both the L2 Understanding level ($p = 0.000 < 0.01$) and the L3 Prediction level ($p = 0.001 < 0.01$), with the game interaction scheme scoring higher than the non-game interaction scheme, as shown in Table 5.4. Among these, the accuracy rate at the L3 level reached 96 % under the game interaction condition, significantly higher than the 88 % under the non-game interaction condition. This indicates that the game interaction mechanism effectively enhances drivers' cognitive modelling of environmental states and their ability to predict future states.

(2) System Usability Analysis.

System usability was assessed using the standardised System Usability Scale (SUS). Participants completed the scale after each scheme test. A one-way analysis of variance (ANOVA) was used to compare usability scores across different schemes. The results showed significant differences in SUS scores between the two interaction schemes ($F = 5.36$, $p = 0.003 < 0.01$), as shown in Table 5.5. In terms of specific scores, the average score for the non-game interaction scheme was 77.18 ($SD = 6.87$), with a system rating of B; the average score for the game interaction scheme was 78.61 ($SD = 7.23$), with a system rating of A. According to the SUS evaluation criteria, both schemes demonstrated good usability, with the game interaction scheme slightly outperforming the non-game interaction scheme in terms of evaluation ratings.

Table 11

Comparison of system availability and reliability and results of variance analysis.

Indicators	Without game interaction	With game interaction	F	P
SUS score	77.18 ± 6.87 (B)	78.61 ± 7.23 (A)	5.36	0.003**
Trust scale	5.36 ± 1.12	5.64 ± 1.43	4.56	0.026*
SA results	78.8 ± 4.56	88.9 ± 5.51	6.094	0.013*

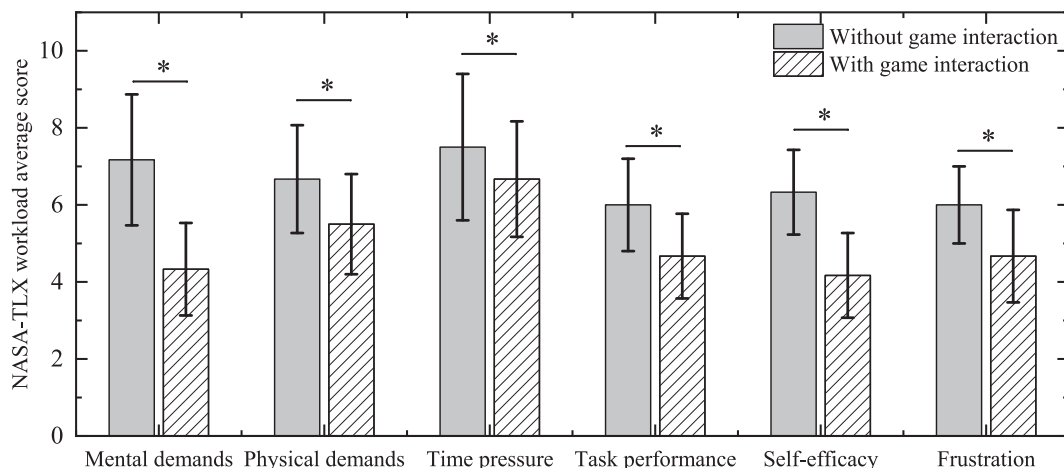


Fig. 16. Average workload scores based on the NASA-TLX scale.

In this study, to comprehensively assess the effectiveness of the multimodal game interaction system in an intelligent cockpit, 10 key evaluation indicators covering three dimensions: physiological indicators, subjective cognition, and task performance, were designed (see Table 7). By comparing the performances under the 8 groups of experimental conditions through a mixed-effects ANOVA model, the study further analyzed the relative contributions of reaction time (N10), gaze stability (N1-N2) and subjective trust (N6) in predicting the effectiveness of the system.

It was found that reaction time (N10) was the most direct objective indicator of system interaction effectiveness. Especially in critical incident scenarios, reaction time was significantly correlated with driving behavior performance and was an important prior variable in predicting system response effectiveness. The attention heat map and attention path (N1, N2) reflect the user's attention distribution and information processing strategy, and are highly correlated with the user's context perception (N4), especially in high load scenarios (e.g., night driving in the city). The stability and concentration of attention can be used as a reference for indirectly assessing the rationality of interface design and the degree of task comprehension. Subjective trust (N6) plays an important role in predicting long-term system acceptance and subjective user satisfaction. The data show that trust is significantly correlated with the user's ability to predict the system (SAGAT-L3), reflecting the fact that trust is the basis of the user's willingness to rely on the system for decision making.

Therefore, when ranked in terms of their ability to predict the overall effectiveness of the system, the importance of the three metrics is roughly as follows: reaction time (N10) > subjective trust (N6) > gaze stability (N1/N2). Among them, reaction time is prioritized for measuring immediate interaction efficiency, subjective trust is used to determine the system's continued willingness to use, and the gaze indicator provides key data support for interface design optimization.

(3) Trustworthiness analysis

Trustworthiness assessment used a 7-point Likert scale with three items (1 = completely untrustworthy, 7 = very trustworthy), with participants completing the assessment separately after each scheme. The results of the normality test indicated that the data did not follow a normal distribution (K-S test, $p = 0.013 < 0.05$), so a non-parametric test (Wilcoxon signed-rank test) was used for inter-solution comparisons. The results indicate that with game interaction can enhance user trust in autonomous driving systems ($p = 0.026 < 0.05$). As shown in Table 12, the average trust score for game-based interactive solutions was 5.64, higher than the 5.36 for non-game-based interactive solutions.

In summary, the game-based interaction prompt mechanism demonstrates significant advantages in enhancing situational awareness (particularly understanding and anticipation), strengthening users' trust in the system, and improving the system's overall usability.

The results of subjective questionnaires and behavioral data show that users participating in multimodal game interactions scored significantly higher than the control group in the dimensions of system trust, entertainment, and engagement, indicating that gamification mechanisms can effectively alleviate the problem of lack of control due to autopilot, and enhance the experience of human-machine collaboration.

According to the analysis results of the SAGAT Situational Awareness

Table 12

Comparison of accuracy rates and results of difference tests for three-layer situation awareness solutions, based on progressive significance two-sided test analysis, with a significance level of 0.05.

SA Level	Without game interaction	With game interaction	Mean Difference	SE	T	P
L1	67.3 %	79.6 %	-10.500	3.630	-2.89	0.004
L2	77.0 %	90.0 %	-29.760	6.020	-4.94	0.000
L3	88.0 %	96.0 %	-19.280	5.823	-3.31	0.001

Scale, SUS Usability Scale, Trust Scale, and behavioral data, users show obvious modal preference characteristics in different task types and driving situations. When dealing with emergencies such as pedestrian intrusion, users preferred to use visual, auditory and haptic modalities concurrently to increase the intensity of warnings, enhance environmental understanding and prediction ability, and thus improve the sense of trust and response accuracy, forming a significant synergistic effect; whereas, in low-risk tasks such as navigation and driving prompts, users preferred the combination of visual and haptic senses, and the presentation of such moderate information could effectively reduce the cognitive load, minimize interference and maintain attention. This moderate information presentation can effectively reduce cognitive load, minimize interference and keep attention; in complex environments such as urban night driving, although the three-modal concurrency improves the system feedback strength, some users are overloaded with feedback information, which is easy to cause distraction and show mild overload phenomenon; in the scene where there is a strong background noise (e.g., construction area) and visual cues are absent, relying on auditory and haptic modalities only can lead to ambiguous orientation, making it difficult to establish spatial references and there is a risk of modal failure or cognitive bias. Therefore, research shows that users are context-sensitive in modal selection, preferring multimodal concurrency for high-risk tasks, simple modalities for routine tasks, and relying on visually dominated combinations in complex environments, which suggests that future interaction systems should construct adaptive multimodal interaction strategies based on context-awareness, and dynamically adjust the modal types and timing of outputs to enhance the safety of driving and the human-machine collaborative experience.

5.4. Discussion

5.4.1. Analysis of overloading or synergistic effects generated by different combinations of interaction modes

In order to explore the overload tasks generated under different combinations of interaction modalities, we systematically compared the performance of different modality combinations under different task types and contexts in formal experiments, and found the following synergy and overload effects, as detailed in Table 13.

(1) Modal synergy effect (synergy). In navigation prompting tasks (e.g., lane change guidance, route selection), the visual + haptic combination shows the optimal synergy effect: the user can quickly recognize the direction, and at the same time, the user can get the operation prompts through slight vibration, which shortens the average reaction time by about 14.7 %, and reduces the user's subjective sense of load. In emergency avoidance tasks (e.g., pedestrian intrusion, sudden braking), the visual + auditory + haptic (tri-modal fusion) is the most effective, and the triple stimulation enhances the concentration of attention, improves the correct recognition rate of the event, and strengthens the user's trust in the system.

(2) Overload. In scenarios with high information interference (e.g.,

Table 13

Analysis of Overload or Synergistic Effects under Different Interaction Modality Combinations.

Task Type	Modality Combination	Observed Effect	Type
Navigation prompts	Vision + Hearing	Fast response, low cognitive load	Synergistic Effect
Emergency events	Vision + Hearing + Touch	High recognition accuracy, increased trust	Synergistic Effect
Urban night driving	All three modalities simultaneously	Good feedback, but information overload	Mild Overload
Noisy background scene	Hearing + Touch (no Vision)	Impaired directional awareness	Overload

urban night driving, complex intersections), simultaneous triggering of the three modes may lead to short-term cognitive overload. Some users reported the phenomenon of “stimulus congestion” and elevated subjective NASA-TLX overload scores in high information content situations. Auditory + Haptic Without visual support, some users misinterpreted the direction of the cue source, resulting in delayed operation, especially in simulated environments with high background noise. For this reason, the system can later incorporate a task-level awareness mechanism that can automatically switch modal combination methods to reduce interference and enhance synergy.

(3) Impact analysis of long-term trust, habituation and desensitization.

Although tri-modal fusion (Vision + Hearing + Touch) significantly enhances users’ attention focus and trust during unexpected events, its long-term use may face the problem of signal “desensitization” or “habituation”. However, its long-term use may face the problem of signal desensitization or habituation. Some subjects indicated in their subjective feedback that frequent triggering of tri-modal alerts, especially in non-critical scenarios, would lead to a gradual decrease in alertness and a weakening of willingness to respond to alerts for critical events. In addition, over-reliance on gamified feedback (e.g., points, virtual currencies) may also trigger a diminishing marginal effect of the reward mechanism, thus weakening the long-term attractiveness and trust building of the system.

From the perspective of game psychology, the lack of adaptive feedback mechanisms in a gamified alert system, such as dynamically adjusting the intensity and frequency of prompts according to the user’s reaction time, eye-movement focus area, or operational performance, may accelerate “stimulus fatigue”. Therefore, future systems should introduce adaptive difficulty adjustment based on user performance and combine positive reinforcement and variable-ratio feedback loop strategies to maintain the novelty and alertness of the alert system. The novelty and alertness of the alert system can be maintained by combining positive reinforcement and variable-ratio feedback loop strategies.

At the same time, we can refer to long-term intervention studies in practical applications to explore the nonlinear relationship between alert frequency, modal combination and user trust, and establish personalized alert models to alleviate the cognitive load and perceptual fatigue caused by high-frequency feedback.

5.4.2. Advantages

Compared with the existing AR and HUD-based alert systems or gamified driving support tools, the proposed method has the following three transcendent advances.

First, in the dimension of interactive experience, this study constructs a multimodal immersive game interaction framework that not only has the information alert function of traditional HUD systems, but also strengthens the user’s emotional engagement and attention through the game mechanism by introducing anthropomorphic virtual characters (e.g., “fire elves”) and the Pokémon-style “collection mechanism”. First, in the dimension of interactive experience, this study constructs a multimodal immersive game interaction framework integrating visual, auditory, and even tactile senses by introducing anthropomorphic virtual characters (e.g., “fire elves”) and a “Pokémon-style” collection mechanism. The change from “prompt-response” to “immersion-modeling” has been realized.

Second, in terms of adaptability and context modeling, we break through the design paradigm of existing HUD systems that are mostly limited to single functions (e.g., navigation, obstacle warning), and based on the three-stage theory of SA, we coupled the risk characteristics of different driving situations (e.g., urban roads, highways, rural roads, etc.) with the user’s behavioral patterns, and proposed an adaptable interactive game plan. By simulating the task response in different unexpected situations, it improves the efficiency of user response to unexpected events in complex driving scenarios and realizes the

preemptive intervention of driving behavior, which is difficult to be realized by the traditional static AR cueing system.

Finally, in terms of empirical evaluation and feedback optimization mechanism, this method not only proposes a gamified interaction scheme, but also systematically verifies its efficacy by means of eye tracking, reaction time analysis, subjective trustworthiness and NASA-TLX cognitive load assessment. Compared with the existing driving support tools that lack multi-dimensional evaluation, the “perception-performance-subjective experience” three-dimensional evaluation framework established in this study has higher reusability and research promotion value, and through the experimental feedback to adjust the complexity of the interface, the optimization of visual elements, and the improvement of incentives, to achieve a data-driven interaction system. Iterative optimization.

In summary, the methodology of this study not only provides a functional enhancement solution for HUD and AR systems, but also realizes a substantial expansion of the methodology in the four dimensions of interaction logic, context awareness, user experience, and system evaluation.

5.4.3. Limitations

Although this study indicates that multimodal gamified interaction offers certain advantages in enhancing user response efficiency and subjective experience in autonomous driving scenarios, it still has limitations.

First, in terms of sample composition, this experiment primarily recruited university students aged 19 to 25 with some experience in AR and gaming. While this approach offers advantages in terms of controlling variables and experimental consistency, the demographic characteristics of the sample are relatively homogeneous and may not fully represent a broader user population (such as older drivers, those with extensive driving experience, or individuals with low technical proficiency). This limits the generalisability and practical applicability of the research conclusions to some extent. Future studies should expand the sample coverage to include participants of different ages, driving backgrounds, and levels of technological acceptance to enhance the representativeness and practical value of the results.

Secondly, in terms of cognitive load measurement, although the study used eye-tracking and the NASA-TLX scale to assess participants’ objective and subjective workload and drew preliminary conclusions, it has not yet explored the specific mechanisms by which multimodal interaction affects attention resources, information processing, and working memory at a more detailed level. To further validate the safety and cognitive adaptability of this interaction method, future research could combine objective indicators such as electroencephalography (EEG), electromyography (EMG), and reaction time variability analysis to establish a more comprehensive cognitive load monitoring framework.

Therefore, the main contribution of this paper lies in providing a validation framework for multi-modal AR game interaction design and preliminarily verifying its effectiveness; however, further expansion and refinement are still needed in terms of sample diversity and cognitive mechanisms.

The AR gamified interaction system used in this study is based on a highly controlled simulated driving environment, and although it helps to systematically test the causal relationship between interface strategies and user responses, we are fully aware of its limitations in terms of ecological validity, and we now provide the following explanations on the related issues:

(1) About the behavioral differences between simulated and real driving.

Studies have shown that in real driving situations, users’ Attention distribution, risk sensitivity, and emotional responses in a real driving situation may differ from those of simulated driving due to factors such as physical involvement (e.g., haptic feedback, road bumps), external disturbances (e.g., noise), and so on. To get as close as possible to the

real experience, we use a high-fidelity driving simulator with steering wheel force (vibration module) feedback, seat vibration, and stereo system, and monitor the user's state through eye tracking and physiological data (e.g., heart rate variability) to enhance the system's ability to capture actual behaviors. In future research, we plan to port the system to a closed-road real-vehicle test environment to further validate its external validity.

(2) On portability under multiple road/weather conditions.

The current experiments are mainly conducted under ideal weather and traffic scenarios, but the proposed AR game mechanism employs a dynamic environment perception and content adaptation module, which is theoretically applicable to complex environments. For example, in low visibility scenarios such as rainy nights, the system can increase the brightness and contrast of the virtual character and superimpose auxiliary sound effects to enhance perceptibility; in highways and congested urban roads, the rhythm of the game task can be adaptively adjusted to avoid excessive interference with driving behavior. In the future version, we will introduce the weather module to simulate rain, fog, night vision, etc., and carry out transferability testing.

Although this system integrates a task adaptation mechanism based on the environmental state: when the system monitors a high-speed or low-visibility scene, it will automatically reduce the frequency and interaction complexity of the game task, suspend the non-core content, and retain only the key situational cues (e.g., risk warnings, navigation markers). This dynamic task scheduling mechanism is designed to avoid further distraction to the driver in high load scenarios, ensuring that the AR elements "enhance attention" rather than "distract". For complex weather environments, the system supports multi-modal fusion of visual enhancements (e.g., high contrast characters), audio alerts, and vibration cues to improve perceptual clarity. For example, in rainy night scenarios, the virtual character will automatically switch to brighter colors and adjust the background lighting to ensure that it is not obscured by the visual noise of the road. Currently, the system is still mainly tested in simulated environments and has not yet been deployed in real highway conditions and bad weather. In the future, we plan to extend the system to real-vehicle testing platforms and introduce driving style recognition and cognitive load sensing modules to intelligently adjust the task tempo and content distribution according to the individual state, so as to improve the generalizability and safety of the system.

(3) About Sensor Delay and AR Occlusion Modeling.

When designing the AR visual overlay logic, this system fully considers the possible occlusion problem and sensor delay in the real environment. First, we dynamically adjust the superimposed position of virtual information through the visual field perception model to avoid blocking the driver's critical field of view in front of the driver; second, in the case of delayed sensor feedback, the system introduces a buffer animation and a gradual emergence mechanism to prevent the sudden change of information from causing jumps in attention or misjudgment. Meanwhile, the future system will integrate an SLAM-based spatial reconstruction module to enhance the robustness and adaptability of virtual-reality fusion.

In addition, considering the user behavior, physiological data (e.g., eye movement trajectory), and real-time recommendation mechanism of the system involved in the multimodal interaction design, the design of the future system has to take into account the challenges of data complexity and user privacy protection. On the one hand, with the distributed collection and processing of interaction data across different devices, it is crucial to make recommendations and personalized adaptations under the premise of protecting user privacy. Related studies have shown that a decentralized smartphone recommendation system that introduces privacy protection can effectively reduce the risk of centralized data leakage and improve the security and trustworthiness of the system [47]. On the other hand, in the analysis of complex data such as user behavior and brain-computer interfaces, the dependence of the model on the complexity of the input data directly affects the

classification accuracy and system robustness. For example, literature [48] proposes a data complexity-based model dependency assessment method for brain MRI images for the identification and classification of brain tumors and Alzheimer's disease, highlighting the risk of generalization of task-critical data under high-dimensional modeling. This research provides important insights on how to construct generalizable and highly trustworthy models in multimodal human-computer interaction.

Therefore, the driving context interaction system based on the fusion of AR and multimodal perception proposed in this paper will further introduce data complexity assessment and privacy-aware mechanisms in future versions to build a more robust, secure and interpretable interaction framework.

6. Conclusion

This paper focuses on multimodal game interaction design within intelligent cockpits, using user research to model autonomous driving scenarios. Based on the situation awareness model, it constructs the information and functional architecture for multimodal games in intelligent cockpits, designing game vision elements and interaction methods. First, the paper explores the development background of intelligent cockpits and autonomous driving, as well as the transformation of the driver's role and user needs, analyzing the current status and trends of multimodal games in vehicles. Second, through in-depth interviews and surveys, the study identifies users' needs for road information perception and analyzes the information elements in typical autonomous driving scenarios, designing corresponding interaction solutions. Finally, an evaluation process was established, using driving simulators and eye-tracking devices to assess the design through subjective scales, task performance, and physiological data, thereby validating the effectiveness and feasibility of the multimodal game design. To improve the scalability, safety, and human-centeredness of multimodal game-based interaction systems and ensure their widespread use in next-generation autonomous driving experiences. Future research could explore how prolonged exposure to gamified cues affects users' trust, alertness, and potential desensitization to critical warnings, especially in high-frequency driving environments. Privacy and data complexity perspectives could also be explored to explore privacy-preserving and decentralized mechanisms for collecting and analyzing multimodal interaction data, as well as to address model dependency and data complexity in behavioral prediction, especially in edge computing environments for smart vehicles.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This research is funded by the scientific research business expenses of central universities; High-level foreign experts program (G2022133031L, G2023133041L); Shenzhen Science and Technology Plan (GJHZ20220913142401002); Project of Sichuan Provincial Key Laboratory of Digital Media Art (23DMAKL01); Category C of Pujiang Project of Shanghai Magnolia Talent Program (23PJC103); China Post-doctoral Science Foundation (2024M752402).

Data availability

The authors do not have permission to share data.

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